

Artistverse
Software Requirement Specification

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		• Activity Diagram • Appendix				
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Chapter 1 Introduction

1.1 Purpose and Scope

The purpose of the software requirement specification document is to present and describe in detail what the system in the Artistverse' project can do. It also includes a set of use cases that describe user interactions that the software must provide. The software specific requirement is also intended to describe the requirement of this project by use case diagram and use case description, define user requirement specification and define the system requirement specification.

1.2 User Characteristics

- Users
 - Songwriters
 - Musicians
 - Beginning Songwriter

Description:

The target users of our software are songwriters, musicians, and beginning songwriters. These three groups have different interests and abilities in creating lyrics, but they share the common goal of making songwriting easier and more enjoyable.

1.3 Acronyms and Definitions

Acronyms

SRS	Software Requirement Specification
URS	User Requirement Specification
UTC	Unit Test Case
STC	System Test Case
UC	Use Case
SRC	Software Requirements Specification

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Definition

Name	Definition
IEEE	Institute for Electrical and Electronics Engineers. Biggest global interest group for engineers of different branches and computer scientists. [1]
Plan	A document series of tasks requires meeting an objective, typically including the associated schedule, budget, resources, organizational description and work breakdown structure. [1]
Feature	Feature Transformation of input parameters to output parameters based on a specified algorithm. It describes the functionality of the product in the language of the product. Used for requirements analysis, design, coding, testing or maintenance. [1]
Project Management	The application of knowledge, skills, tools, and techniques to project activities in order to meet or exceed stakeholder needs and expectations from a project. [1]
Project Plan	A formal, approved document used to guide both project execution and project control. The primary uses of the project plan are to document planning assumptions and the decision, to facilitate communication among stakeholders, and to document approved scope, cost, and schedule baseline. [1]
Risk	An uncertain event or condition that, if it occurs, has a positive or negative effect on a project's objectives. It is a function of the probability of occurrence of a given threat's occurrence. [1]
Risk Management	The systematic application of management policies, procedures and practices to the tasks of identifying, analyzing, evaluating, treating and monitoring risk. [1]
System Testing	Testing conducted on a complete and integrated system for evaluate the system's compliance with its specified requirements. [1]
Unit Test	A test of individual programs or modules in order to remove a design or programming errors. [1]

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Traceability	The ability to trace the history, application or location of an item or activity, or work products or activities, by means of recorded identification. The establishment and maintenance of relationships between such items. Horizontal traceability describes the relationship between work products of the same type (e.g. Customer requirements). Vertical traceability describes the relationship between work products, which build or derived from each other (e.g., from customer requirements to qualification test cases). Bidirectional traceability allows to directly following relationships in both directions. [1]
Use Case	A use case is a description of a specific interaction between a system and its users to accomplish a particular goal or task. It outlines the sequence of steps or actions that a user performs in the system and describes the system's responses to those actions. Use cases help to identify and define the functional requirements of a system from the perspective of its users.
Activity Diagram	Activity diagram is another important behavioral diagram in UML diagram to describe dynamic aspects of the system. Activity diagram is essentially an advanced version of flow chart that modeling the flow from one activity to another activity.

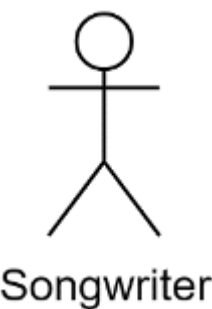
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Chapter 2 Specification Requirement

2.1 Progress I

2.1.1 Use Case Scenario

2.1.1.1 Actor

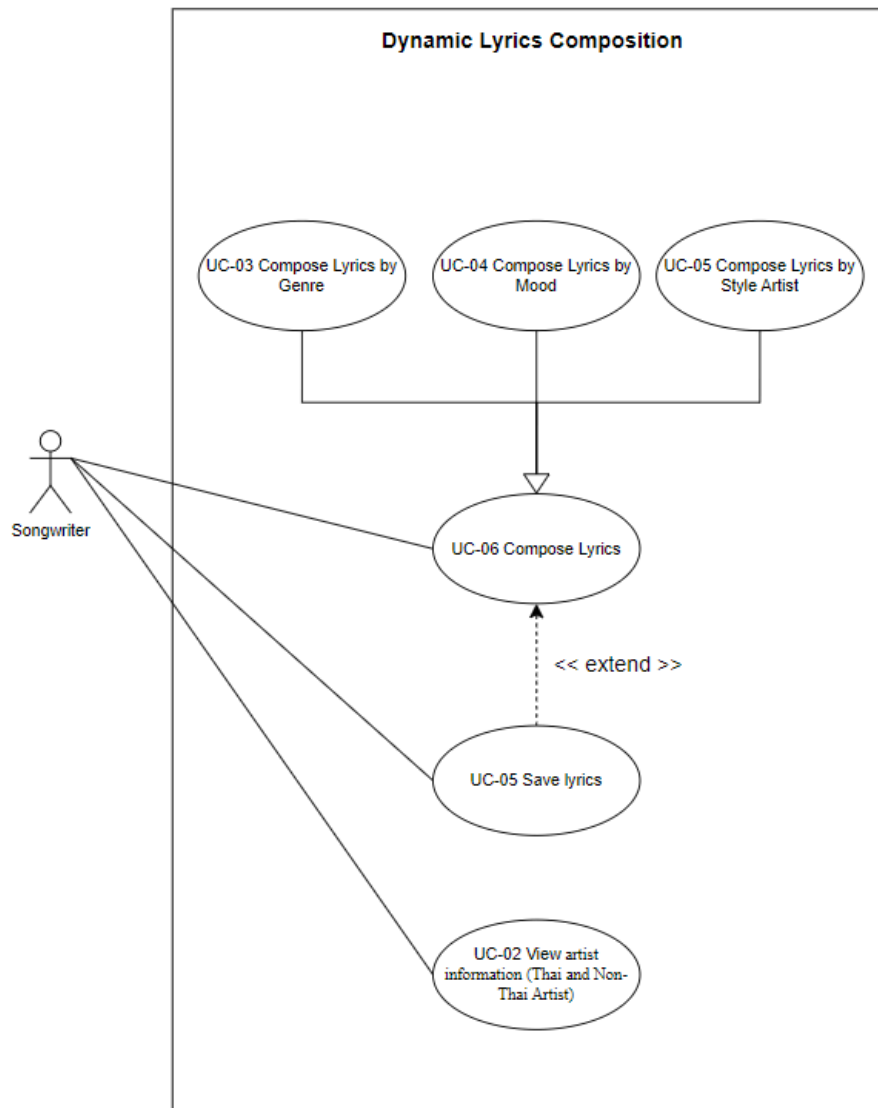


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2.1.1.2 Use Case Diagram

Feature 1: Dynamic lyrics Composition

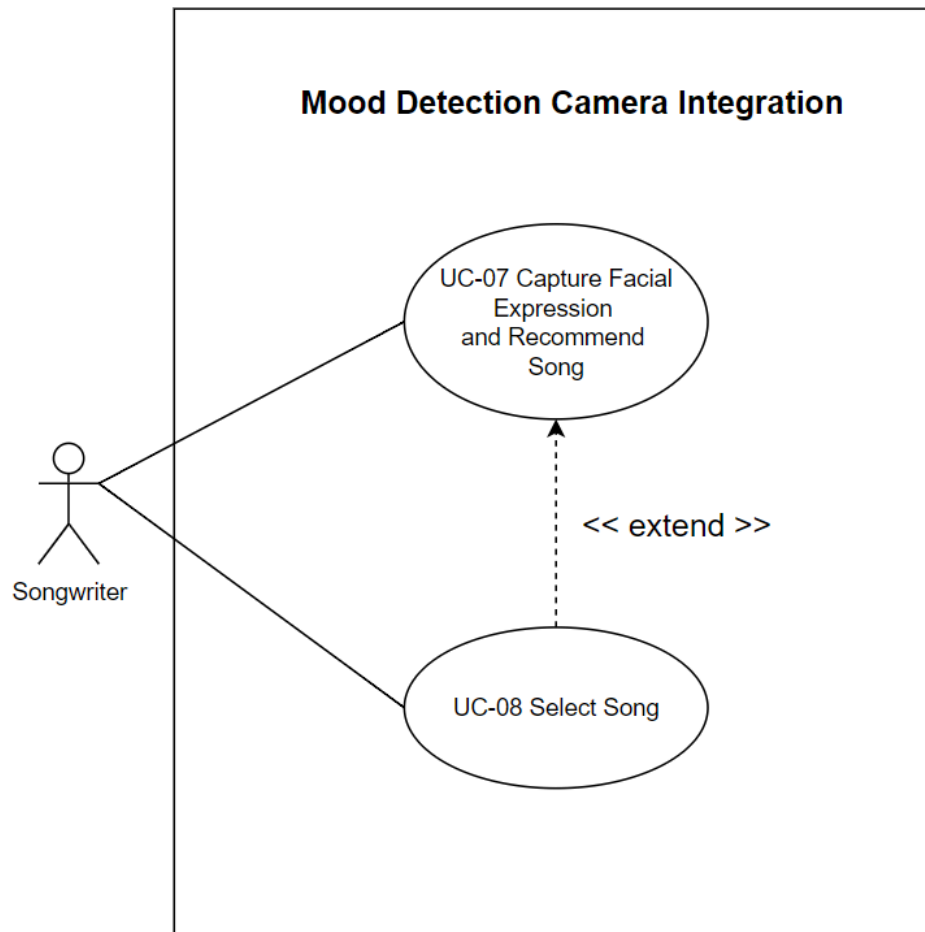
This use case diagram describes the part of this function work. In UC-01, songwriters can access the compose lyrics and the lyrics provide an option to the songwriter to choose in UC-02,UC-03,UC-04 also when the songwriter completes the form that is selected by the songwriter. System provides the save lyrics function to tell the songwriter that the song can be saved to their device.



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Feature 2: Mood Detection Camera Integration

This use case diagram describes each part of this function. Firstly, Songwriters can access the Capture facial expression and receive the recommended song by capturing the facial expression and the system will display the song based on songwriter facial expression. In UC-07. after the Songwriter received the recommended song from the system. Songwriters can also select songs to listen to on the Spotify platform.

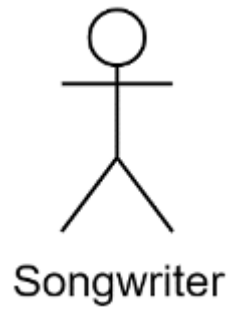


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2.2 Progress II

2.2.1 Use Case Scenario

2.2.1.1 Actor

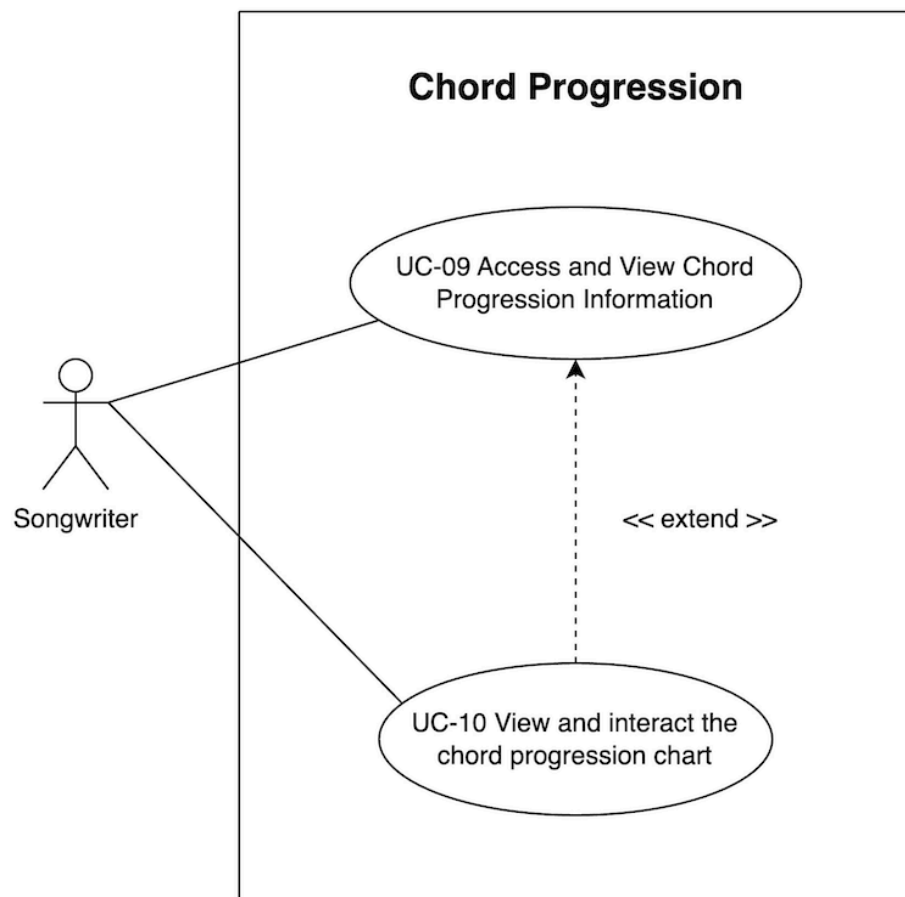


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2.2.1.2 Use Case Diagram

Feature 3: Chord Progression

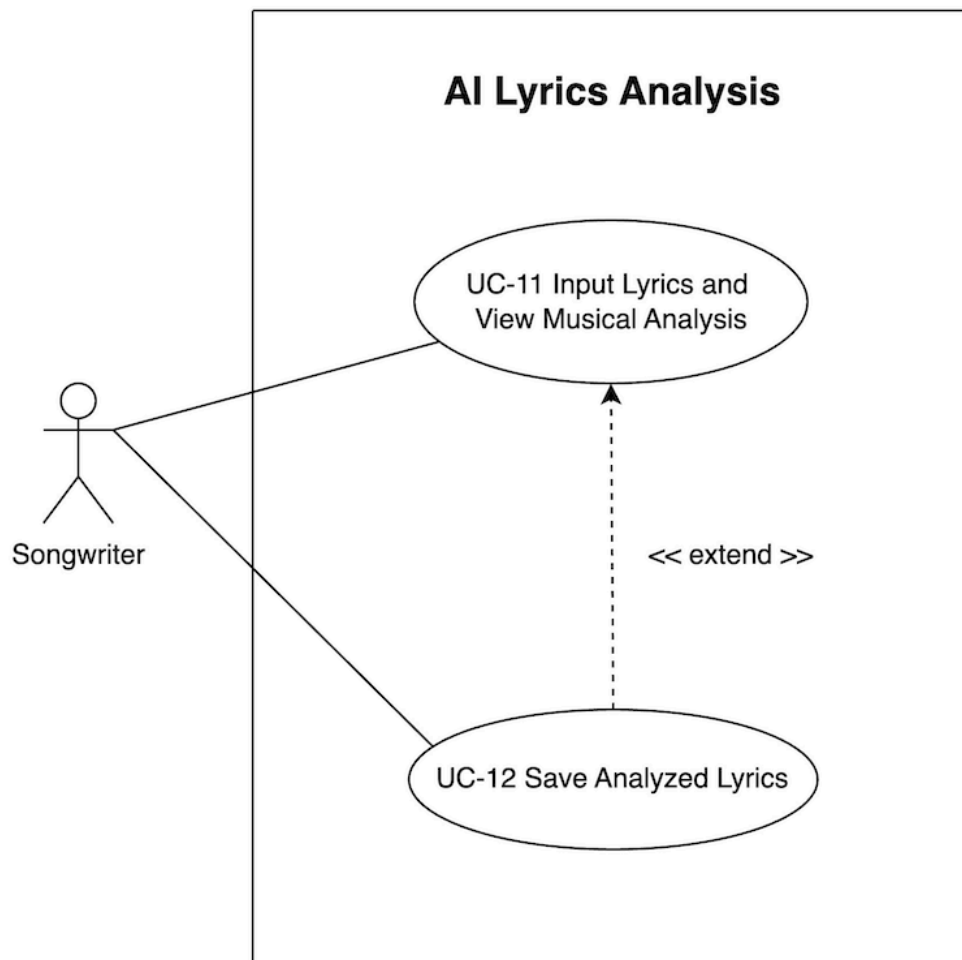
This use case allows the songwriter to access and view detailed chord progression information. It represents the main entry point for interacting with the chord progression features. This use case extends UC-10, meaning that when the songwriter accesses chord progression information, they can also choose to view and interact with the chord progression chart.



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Feature 4: AI Lyrics Analysis

This use case allows the songwriter to input lyrics and receive a musical analysis from an AI Analysis page. The songwriter can view how the lyrics align with musical elements such as melody, rhythm, and harmony. This use case extends UC-12, indicating that after inputting lyrics and viewing the analysis, the songwriter can save the lyrics after they have been analyzed by the AI system. It extends UC-11, meaning that saving the lyrics is an optional extension after viewing the musical analysis.



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2.1.2 User Requirement Specification

Feature#1: Dynamic Lyric Composition

URS-01: The songwriter can access lyric-generation options. (Genre, Mood, and Style Artist)

URS-02: The songwriter can view artist information. (Thai and Non-Thai Artists)

URS-03: The songwriter can compose lyrics by Genre. (pop rock, country R&B, hip-hop, folk, indie, electronic jazz, blues reggae, classical, alternative, gospel, and punk)

URS-04: The songwriter can compose lyrics by Mood. (sad, angry, surprised, fearful, and happy)

URS-05: The songwriter can compose lyrics from Style Artist. (Taylor Swift, Billie Eilish, Ed Sheeran, Justin Bieber, The Weeknd, Maroon 5, Olivia Rodrigo, The 1975, Joji, Keshi, Potato, Palmy, Big Ass, Lula, Lipta, Scrubb, Stamp, Getsunova, and Da Endorphine.)

URS-06: The Songwriter can save lyrics after composing lyrics.

Feature#2 Mood Detection Camera Integration

URS-07: The songwriter can receive recommended songs from Spotify according to the mood from facial expressions mood

URS-08: The songwriter can listen to songs recommended in playlists on Spotify.

Feature #3: Chord Progression

URS-09: The songwriter can access the "Chord Progression" page to view information, selecting from the navigation options provided in SRS-02.

URS-10: The songwriter can view the chord progression chart after accessing the "Chord Progression" page.

Feature #4: AI Lyrics Analysis

URS-11: The songwriter can input lyrics into a text field and view a detailed musical analysis based on the lyrics input, selecting from the navigation options provided in SRS-02.

URS-12: The songwriter can save information as a .txt file on their local computer after analyzing the input lyrics.

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2.1.3 Use Case Description and Activity Diagram

2.1.3.1 UCD-01

URS-01: The songwriter can access lyric-generation options. (Genre, Mood, and Style Artist)

- SRS-01: The system shall display the "navbar-toggler-icon" button on every page's top right bar.
- SRS-02: The system shall display a drop-down menu bar with buttons for "Homepage", "Lyric Generator", "Mood Camera", "Chord Progression" and "AI-Analysis." Clicking on each button will take the user to the respective page.
- SRS-03: After the "Lyric Generator" button on the navbar is clicked, The system shall display a lyrics generator page [UI-03] with the "let's generate" button to start generating lyrics.
- SRS-04: After the "let's generate" button on UI-03 is clicked, The system shall display options for composing lyrics, including "Lyric Generator by Genre", "Lyric Generator by Mood" and "Lyric Generator by Style Artist" after the "let's generate" button is clicked.

Use Case ID	UC-01		
Use Case name	Access Lyric Generation Options		
Created By	NS, WS	Last Update By	NS, WS
Data create	10/05/2024	Last revision	29/05/2024
Actors	Songwriter		
Description	This use case describes the process by which a songwriter navigates to the lyrics generator page and accesses various options for composing lyrics, such as by genre, mood, and style artist.		
Trigger	Songwriter click the “navbar-toggler-icon” button in the top right bar of the website and select Lyric Generator.		
Preconditions	"navbar-toggler-icon" button on every page present on the webpage. [UI-02]		
Use Case Input Specification			
Input	Type	Constraint	Example
-	-	-	-
Post conditions	The songwriter has access to the Lyric Generator by Genre page [UI-04].		
Normal flows	User	System	

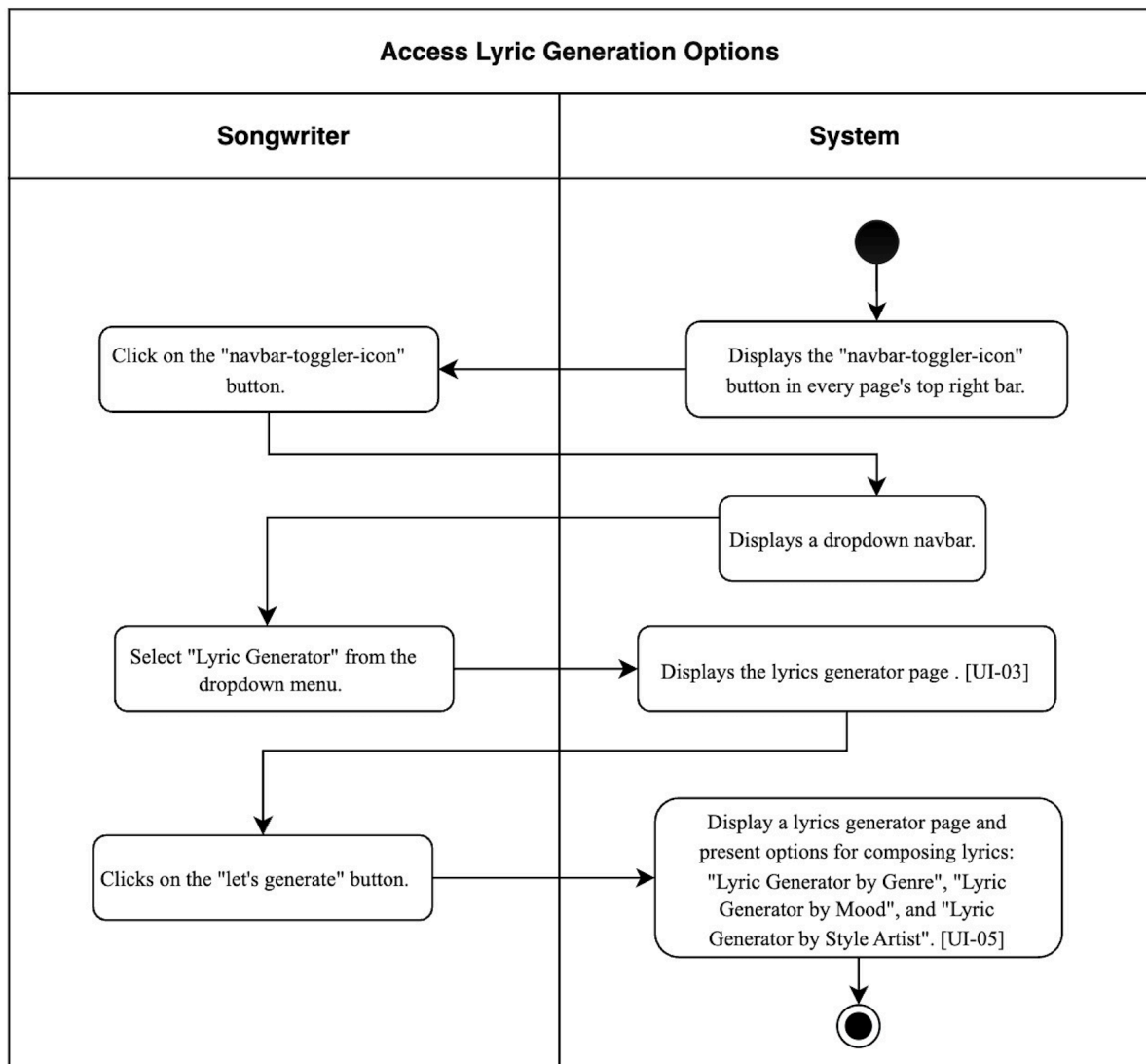
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	2. Click on the "navbar-toggler-icon" button. 4. Select "Lyric Generator" from the dropdown menu. 6. Click on the button "Let's Generate"	1. Displays the "navbar-toggler-icon" button in every page's top right bar. 3. displays a dropdown navbar. 5. Displays the lyrics generator page. [UI-03] 7. Display a lyrics generator page and present options for composing lyrics: "Lyric Generator by Genre", "Lyric Generator by Mood", and "Lyric Generator by Style Artist". [UI-04]
Alternate Flow	-	
Exception Flow	-	
Assumption	The songwriter has an internet connection	

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AD-01: Access Lyric Generation Options

The activity diagram shows the steps to using the lyric generator, starting with accessing the drop-down menu on the website and selecting the lyric generator page, which gives users the option to choose how to generate lyrics based on Genre, Mood, or Style Artist, to make it easier for songwriters to create lyrics.



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2.1.3.2 UCD-02

URS-02: The songwriter can view artist information. (Thai and Non-Thai Artists)

- SRS-05: After the "Show Thai Artists" button on UI-03 is clicked, the system shall display the picture and name of the Thai artist along with the "Show Non-Thai Artists" button.
- SRS-06: After the "Show Non-Thai Artists" button on UI-03-1 is clicked, the system shall display the picture and name of the non-Thai artist along with the "Show Thai Artists" button.
- SRS-07: On the UI-03, the system shall provide a scroll bar to enable the songwriter to navigate through additional non-Thai artist information beyond the initial viewable area.
- SRS-08: On the UI-03-1, the system shall provide a scroll bar to enable the songwriter to navigate through additional Thai artist information beyond the initial viewable area.

Use Case ID	UC-02		
Use Case name	View Artist Information		
Created By	NS, WS	Last Update By	NS, WS
Data create	10/05/2024	Last revision	29/05/2024
Actors	Songwriter		
Description	This use case describes the process by which a songwriter views and switches between Thai and non-Thai artists on the lyric generator page.		
Trigger	Songwriter clicks the "Show Thai Artists" button [UI-03] or "Show Non-Thai Artists" button [UI-03-1].		
Preconditions	The songwriter is on the lyrics generator page with artist information displayed. [UI-03]		
Use Case Input Specification			
Input	Type	Constraint	Example
-	-	-	-
Post conditions	The system displays the selected artist information (Thai or non-Thai) with a button to switch back.		
Normal flows	User	System	

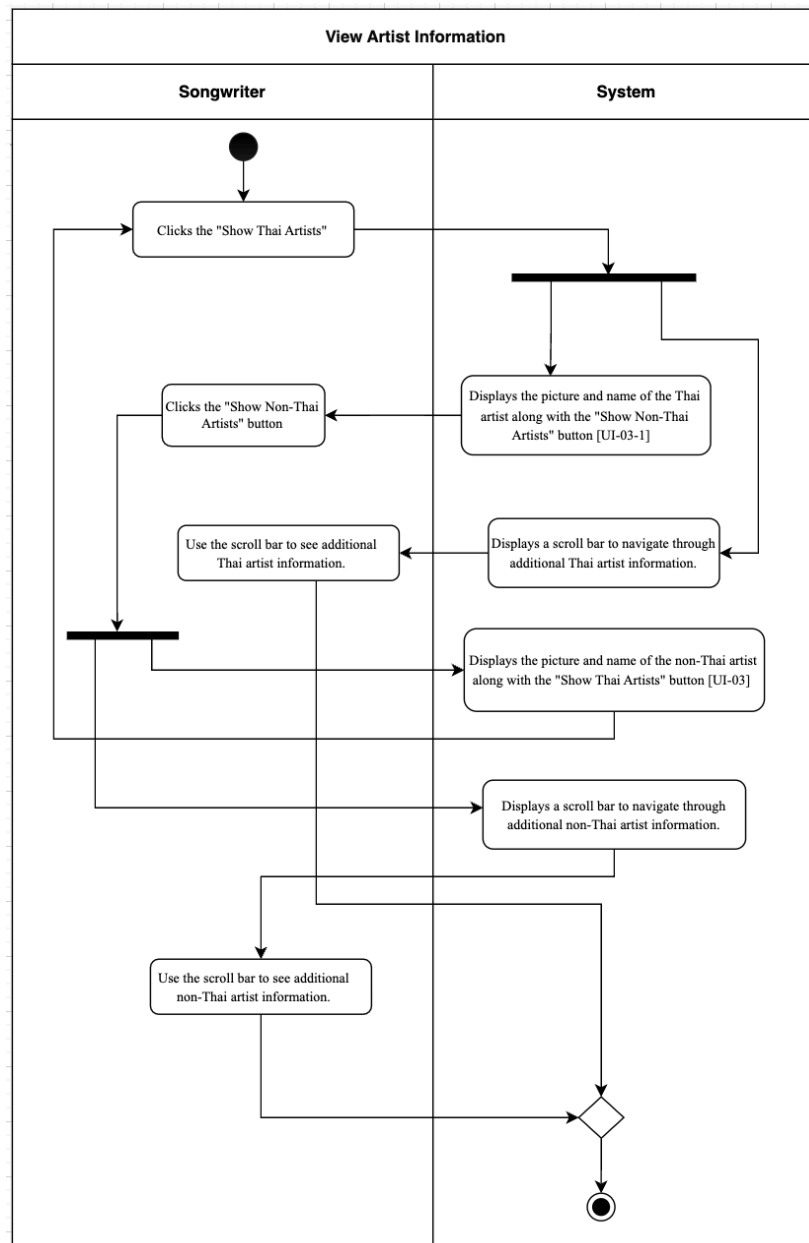
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	1. Click the "Show Thai Artists" button. 4. Use the scroll bar to see additional Thai artist information. 5. Click the "Show Non-Thai Artists" button. 6. Use the scroll bar to see additional non-Thai artist information.	2. Displays Thai artist information along with the "Show Non-Thai Artists" button. [UI-03-1] 3. Displays a scroll bar to navigate through additional Thai artist information. 5. Displays the picture and name of the non-Thai artist along with the "Show Thai Artists" button. [UI-03] 6. Displays a scroll bar to navigate through additional non-Thai artist information.
Alternate Flow	-	
Exception Flow	-	
Assumption	The songwriter has an internet connection	

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AD-02: View Artist Information

The activity diagram shows the steps to view and switch between Thai and non-Thai artist information on the lyric generator page.



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2.1.3.3 UCD-03

URS-03: The songwriter can compose lyrics by Genre. (pop rock, country R&B, hip-hop, folk, indie, electronic jazz, blues reggae, classical, alternative, gospel, and punk).

- SRS-09: The system shall display a lyrics generator page with options for "Lyric Generator by Genre," "Lyric Generator by Mood," and "Lyric Generator by Style Artist."
- SRS-10: The system shall provide forms to fill in the genre information, including lyric topic, genre, and language.
- SRS-11: The system shall generate lyrics based on the input provided by the user.
- SRS-12: If the songwriter attempts to proceed without completing all required genre information input fields, the system shall display an error message indicating the missing fields. The error message consists of "Please input a lyric topic.", "Please select genre.", "Please select a language. [UI-04-1]
- SRS-13: If there is no internet connection, the system shall display an error message "Unable to compose lyrics." [UI-04-4]
- SRS-14: The system shall allow the songwriter to generate new lyrics by filling in the genre information form again on the Lyric Generator by Genre page. [UI-04-2]
- SRS-15: The system shall display the created lyrics to the songwriter based on the songwriter selected.

Use Case ID	UC-03		
Use Case name	Compose Lyrics by Genre		
Created By	NS, WS	Last Update By	NS, WS
Data create	15/05/2024	Last revision	29/05/2024
Actors	Songwriter		
Description	This use case describes generating lyrics based on a selected genre.		
Trigger	The songwriter selects the "Lyric Generator by Genre" option on the lyrics generator page. [UI-04]		
Preconditions	UC-01		
Use Case Input Specification			
Input	Type	Constraint	Example
Lyric topic	String	Required	“My love”
Genre	Dropdown selection	Select One Genre: pop, rock, country, R&B, hip hop, folk, indie, electronic jazz, blues,	“pop”

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		reggae, classical, alternative, gospel, punk	
Language	Dropdown selection	Select One Language: English, Thai, Japanese, Chinese	“English”
Post conditions	The generated lyrics are displayed to the songwriter.		
Normal flows	User	System	
	2. Select "Lyric Generator by Genre". [UI-04] 4. The songwriter fills in information in the form [UI-04-2] 5. Click on the "Generate" button to generate lyrics. [UI-04-2]	1. Display a lyrics generator page and present options for composing lyrics: "Lyric Generator by Genre", "Lyric Generator by Mood", and "Lyric Generator by Style Artist". [UI-04] 3. Provide forms to fill in the genre information, including the lyric topic, genre, and language. [UI-04] 6. Check if all required input fields are completed. <i>(6A: If the songwriter attempts to proceed without completing all required input fields.)</i> 7. The system displays the created lyrics to the songwriter based on the songwriter selected. [UI-04-3] <i>(7A: Unable to displays the lyrics causing by no internet connection)</i> <i>(7B: If the songwriter wants to generate new lyrics by fill information in the form.)</i>	

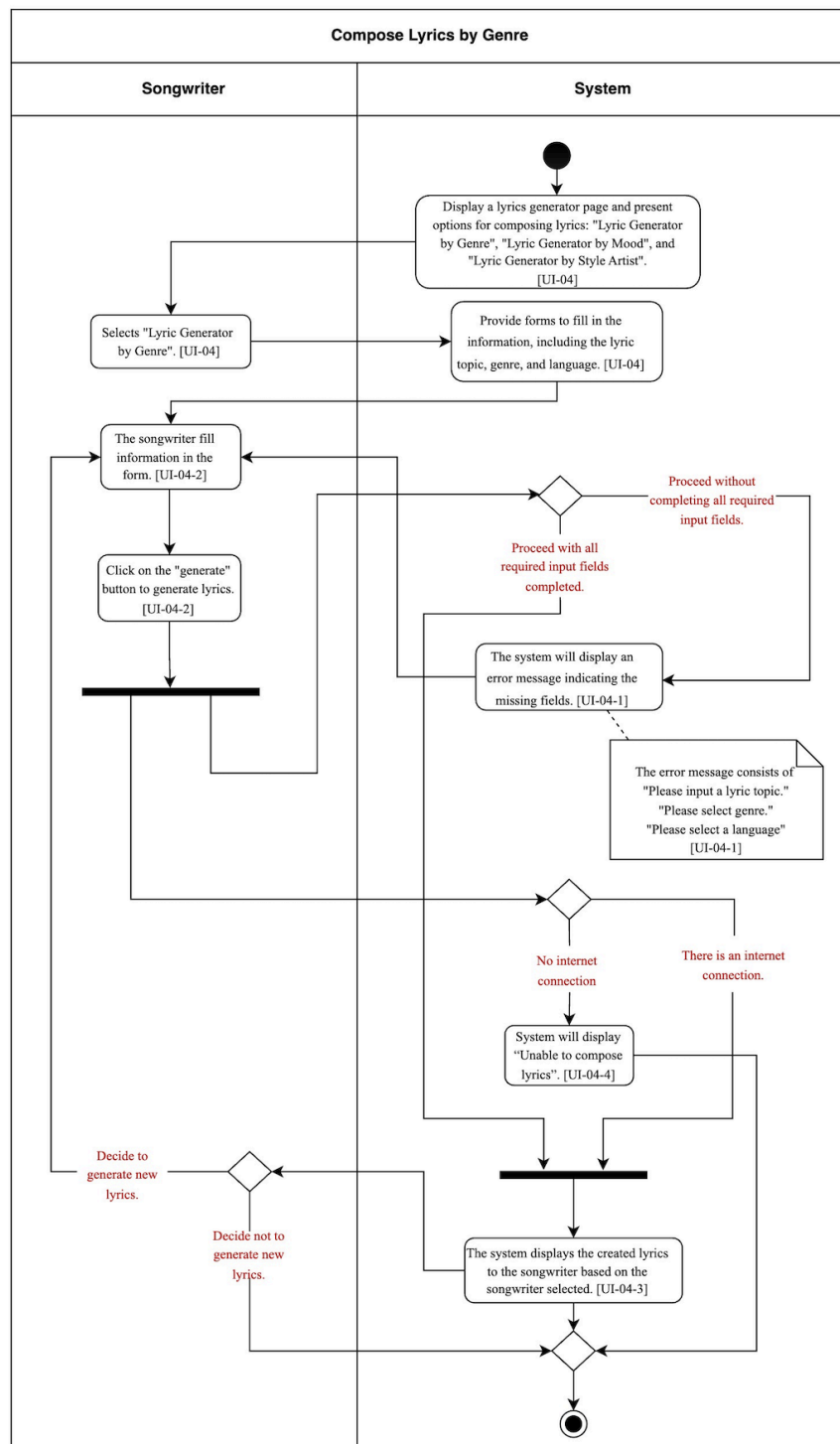
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Alternate Flow	[6A: If the songwriter attempts to proceed without completing all required input fields.] 1. The system will display an error message indicating the missing fields. The error message consists of "Please input a lyric topic.", "Please select genre.", "Please select a language. [UI-04-1] 2. Go to normal flow 4 [7B: If the songwriter wants to generate new lyrics by fill information in the form.] 1. Go to normal flow 4.
Exception Flow	[7A: Unable to displays the lyrics causing by no internet connection] 1. The system will display “Unable to compose lyrics” [UI-04-4] 2. The use case ends.
Assumption	The songwriter has an internet connection

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AD-03: Compose Lyrics by Genre

Activity Diagram for Compose Lyrics by Genre, which explains the steps. Between a songwriter and a system for creating Genre lyrics.



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2.1.3.4 UCD-04: Compose Lyrics by Mood

URS-04: The songwriter can compose lyrics by Mood. (sad, angry, surprised, fearful, and happy)

- SRS-16: The system shall display a lyrics generator page with options for "Lyric Generator by Genre," "Lyric Generator by Mood," and "Lyric Generator by Style Artist."
- SRS-17: The system shall provide forms to fill in the mood information, including lyric topic, mood, and language.
- SRS-18: The system shall generate lyrics based on the input provided by the user.
- SRS-19: If the songwriter attempts to proceed without completing all required mood information input fields, the system shall display an error message indicating the missing fields. The error message consists of "Please input a lyric topic.", "Please select mood.", "Please select a language."
- SRS-20: If there is no internet connection, the system shall display an error message "Unable to compose lyrics." [UI-05-3]
- SRS-21: The system shall allow the songwriter to generate new lyrics by filling in the mood information form again on the Lyric Generator by Mood page. [UI-05-1]
- SRS-22: The system shall display the created lyrics to the songwriter based on the songwriter selected.

Use Case ID	UC-04		
Use Case name	Compose Lyrics by Mood		
Created By	NS, WS	Last Update By	NS, WS
Data create	15/05/2024	Last revision	29/05/2024
Actors	Songwriter		
Description	This use case describes the process of generating lyrics based on a selected mood.		
Trigger	The songwriter selects the "Lyric Generator by Mood" option on the lyrics generator page. [UI-05]		
Preconditions	UC-01		
Use Case Input Specification			
Input	Type	Constraint	Example
Lyric topic	String	Required	“My love”
Mood	Dropdown selection	Select One Mood: happy, sad, romantic, inspirational, energetic,	“Sad”

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		chill, hopeful, reflective, mysterious, dramatic, nostalgic, playful, confident, dreamy, empowering	
Language	Dropdown selection	Select One Language: English, Thai, Japanese, Chinese	“English”
Post conditions	The generated lyrics are displayed to the songwriter.		
Normal flows	User	System	
	2. Select "Lyric Generator by Mood". [UI-05] 4. The songwriter fills in information in the form. [UI-05-1] 5. Click on the "Generate" button to generate lyrics. [UI-05-1]	1. Display a lyrics generator page and present options for composing lyrics: "Lyric Generator by Genre", "Lyric Generator by Mood", and "Lyric Generator by Style Artist". [UI-04] 3. Provide forms to fill in the mood information, including the lyric topic, mood, and language. [UI-05] 6. Check if all required input fields are completed. (6A: If the songwriter attempts to proceed without completing all required input fields.) 7. The system displays the created lyrics to the songwriter based on the songwriter selected. [UI-05-4] (7A: Unable to displays the lyrics causing by no internet connection)	

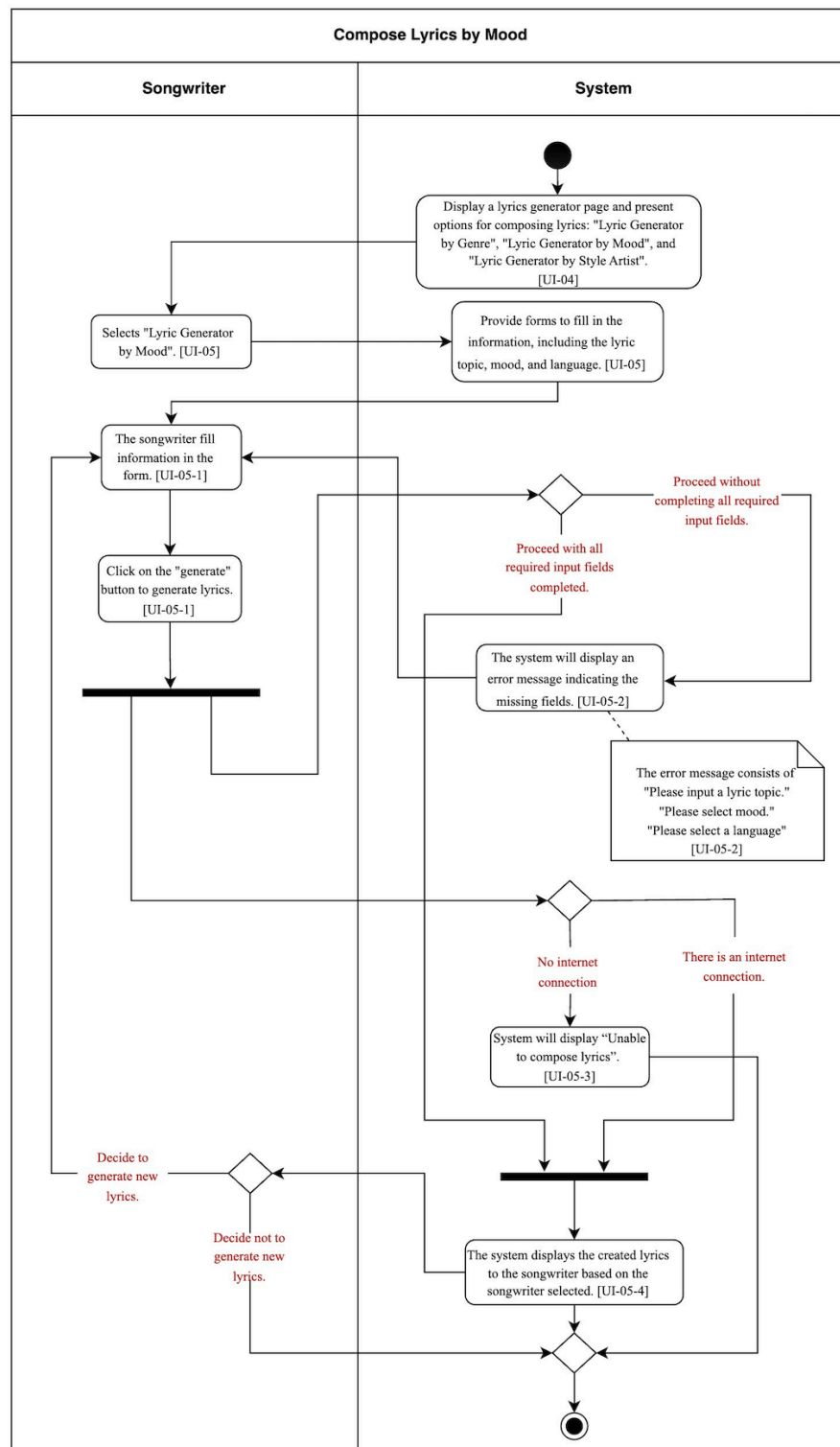
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		(7B: If the songwriter wants to generate new lyrics by fill information in the form.)
Alternate Flow	[6A: If the songwriter attempts to proceed without completing all required input fields.] 1. The system will display an error message indicating the missing fields. The error message consists of "Please input a lyric topic.", "Please select mood.", "Please select a language. [UI-05-2] 2. Go to normal flow 4 [7B: If the songwriter wants to generate new lyrics by fill information in the form.] 1. Go to normal flow 4.	
Exception Flow	[7A: Unable to displays the lyrics causing by no internet connection] 1. System will display “Unable to compose lyrics” [UI-05-3] 2. The use case ends.	
Assumption	The songwriter has an internet connection	

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AD-04: Compose Lyrics by Mood

Activity Diagram for Compose Lyrics by Mood, which explains the steps. Between a songwriter and a system for creating emotional lyrics.



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2.1.3.5 UCD-05 : Compose Lyrics by Style Artist

URS-05: The songwriter can compose lyrics from Style Artist. (Taylor Swift, Billie Eilish, Ed Sheeran, Justin Bieber, The Weeknd, Maroon 5, Olivia Rodrigo, The 1975, Joji, Keshi, Potato, Palmy, Big Ass, Lula, Lipta, Scrubb, Stamp, Getsunova, and Da Endorphine.)

- SRS-23: The system shall display a lyrics generator page with options for "Lyric Generator by Genre," "Lyric Generator by Mood," and "Lyric Generator by Style Artist."
- SRS-24: The system shall provide forms to fill in the style artist information, including lyric topic, artist, and language.
- SRS-25: The system shall generate lyrics based on the input provided by the user.
- SRS-26: If the songwriter attempts to proceed without completing all required genre information input fields, the system shall display an error message indicating the missing fields. The error message consists of "Please input a lyric topic.", "Please select artist.", "Please select a language. [UI-06-1]
- SRS-27: If there is no internet connection, the system shall display an error message "Unable to compose lyrics." [UI-06-3]
- SRS-28: The system shall allow the songwriter to generate new lyrics by filling in the genre information form again on the Lyric Generator by Genre page. [UI-06-2]
- SRS-29: The system shall display the created lyrics to the songwriter based on the songwriter selected.

Use Case ID	UC-05		
Use Case name	Compose Lyrics by Style Artist		
Created By	NS, WS	Last Update By	NS, WS
Data create	15/05/2024	Last revision	29/05/2024
Actors	Songwriter		
Description	This use case describes the process of generating lyrics based on a selected style artist.		
Trigger	The songwriter selects the "Lyric Generator by Style Artist" option on the lyrics generator page. [UI-06]		
Preconditions	UC-01		
Use Case Input Specification			
Input	Type	Constraint	Example
Lyric topic	String	Required	“Sad story”

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Artist	Dropdown selection	Select One Artist: Taylor Swift, Billie Eilish, Ed Sheeran, Justin Bieber, The Weeknd, Maroon 5, Olivia Rodrigo, The 1975, Joji, Keshi, Potato, Palmy, Big Ass, Lula, Lipta, Scrubb, Stamp, Getsunova, Da Endorphine	“Taylor Swift”
Language	Dropdown selection	Select One Language: English, Thai, Japanese, Chinese	“English”
Post conditions	The generated lyrics are displayed to the songwriter.		
Normal flows	User	System	

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	2. Select "Lyric Generator by Style Artist". [UI-06] 4. The songwriter fills in information in the form [UI-06-2] 5. Click on the "Generate" button to generate lyrics. [UI-06-2]	1. Display a lyrics generator page and present options for composing lyrics: "Lyric Generator by Genre", "Lyric Generator by Mood", and "Lyric Generator by Style Artist". [UI-04] 3. Provide forms to fill in the style artist information, including the lyric topic, artist, and language. [UI-06] 6. Check if all required input fields are completed. <i>(6A: If the songwriter attempts to proceed without completing all required input fields.)</i> 7. The system displays the created lyrics to the songwriter based on the songwriter selected. [UI-06-4] <i>(7A: Unable to displays the lyrics causing by no internet connection)</i> <i>(7B: If the songwriter wants to generate new lyrics by fill information in the form.)</i>
Alternate Flow	[6A: If the songwriter attempts to proceed without completing all required input fields.] 1. The system will display an error message indicating the missing fields. The error message consists of "Please input a lyric topic.", "Please select artist.", "Please select a language. [UI-06-1] 2. Go to normal flow 4	

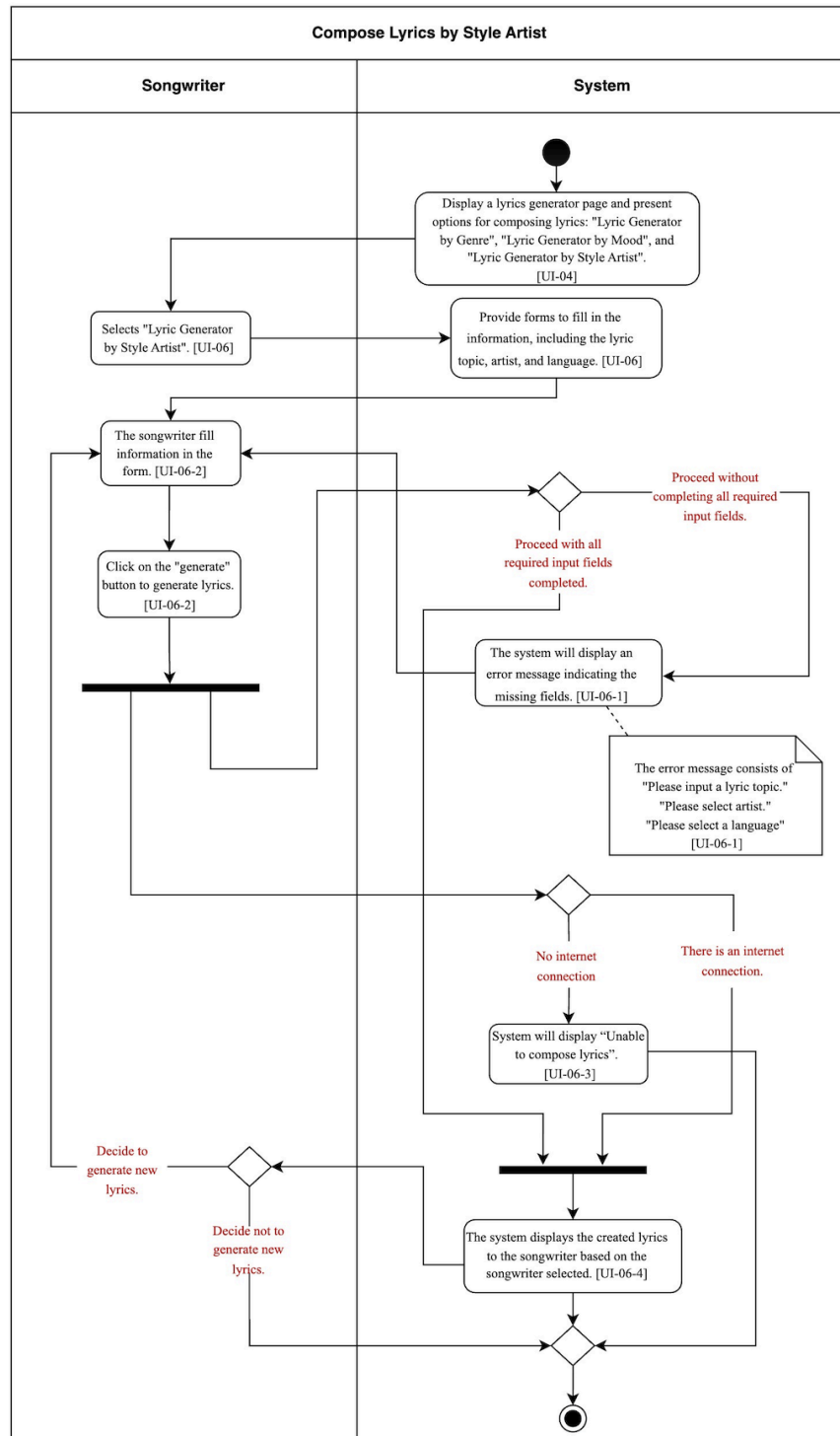
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	[7B: If the songwriter wants to generate new lyrics by fill information in the form.] 1. Go to normal flow 4.
Exception Flow	[7A: Unable to displays the lyrics causing by no internet connection] 1. System will display “Unable to compose lyrics” [UI-06-3] 2. The use case ends.
Assumption	The songwriter has an internet connection

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AD-05: Compose Lyrics by Style Artist

This activity diagram shows how to compose lyrics using a style artist feature in a lyrics generation system. It describes the steps between the songwriter and the system to create lyrics in the style of a chosen artist.



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2.1.3.6 UCD-06 : Save lyrics

URS-06: The songwriter can save lyrics after composing lyrics.

- SRS-30: The system shall provide a "Save" button that allows the songwriter to save the created lyrics. [UI-04-3, UI-05-4, UI-06-4]
- SRS-31: the system shall save the lyrics to the songwriter's local computer in a .txt file format. [UI-07]
- SRS-32: If there is no internet connection, the system shall display an error message "Unable to save lyrics." [UI-07-1]

Use Case ID	UC-06		
Use Case name	Save lyrics		
Created By	NS, WS	Last Update By	NS, WS
Data create	11/05/2024	Last revision	29/05/2024
Actors	Songwriter		
Description	This use case describes the process for saving generated lyrics to the songwriter's local computer.		
Trigger	The songwriter clicks the "Save" button after the lyrics have been generated and displayed in the UC-02, UC-03, or UC-04.		
Preconditions	UC-02, UC-03, or UC-04.		
Use Case Input Specification			
Input	Type	Constraint	Example
Save action	button	User-initiated	Clicking "Save"
Post conditions	The lyrics are saved to the songwriter's local computer in a .txt file.		
Normal flows	User	System	

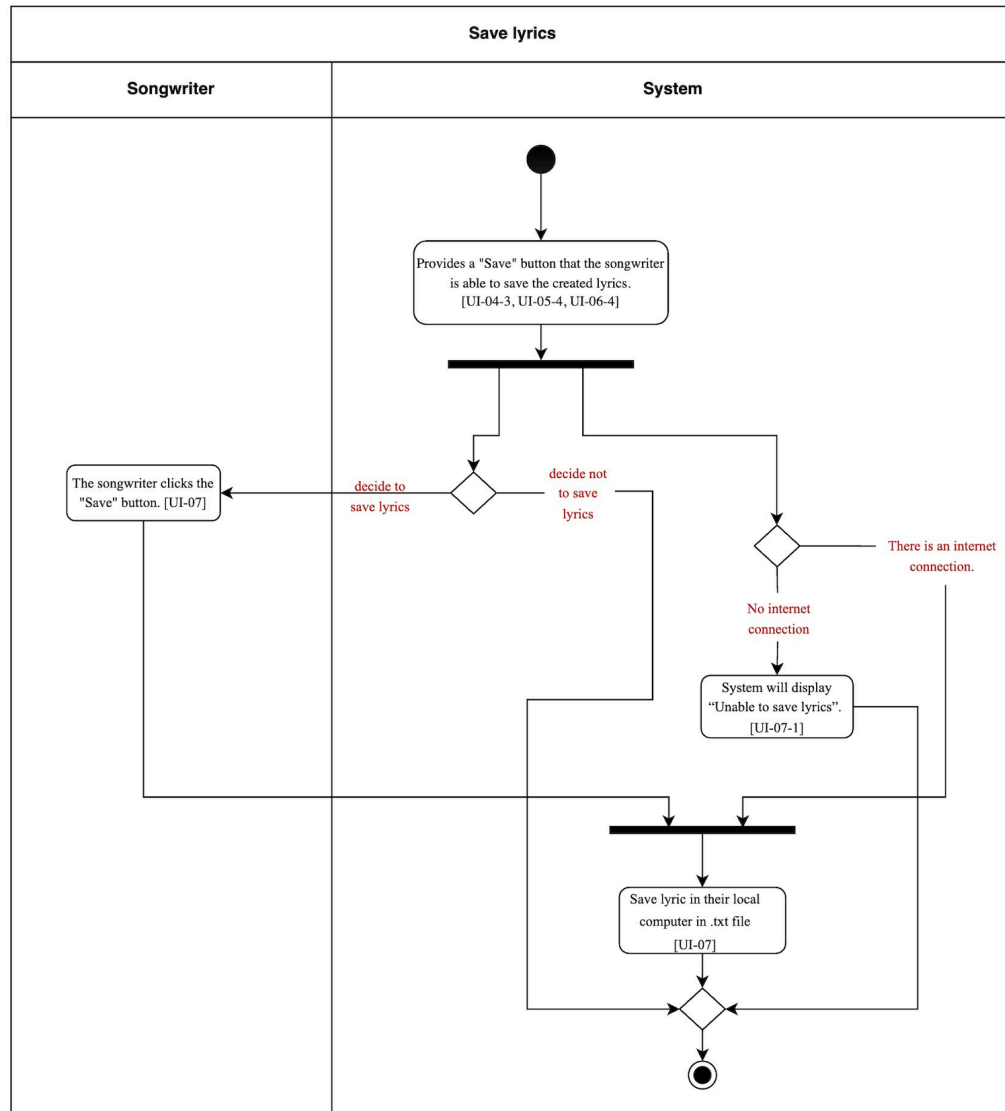
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	2. The songwriter clicks the "Save" button.	1. Provides a "Save" button that the songwriter can save the created lyrics. [UI-04-3, UI-05-4, UI-06-4] 3. Save lyric in their local computer in .txt file [UI-07] <i>(3A: Unable to save the lyrics causing by no internet connection)</i>
Alternate Flow	-	
Exception Flow	[3A: Unable to save the lyrics causing by no internet connection.] 1. The system will display “Unable to save lyrics” [UI-07-1] 2. The use case ends.	
Assumption	• The songwriter has already generated lyrics using one of the lyric generation options available in the system. • The songwriter has an internet connection	

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AD-06: Save lyrics

The Save lyric activity diagram will show the activity flow of the songwriter can save lyrics after composing lyrics.



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2.1.3.7 UCD-07

URS-07: Songwriter can capture facial expressions to detect mood and received recommended song

- SRS-33: The system shall display a "Camera Mood" page [UI-08]
- SRS-34: The system will display a message to ask permission for the camera access including allow and deny.
- SRS-35: The system shall display the camera view. [UI-08-1]
- SRS-36: The system shall receive the facial expression of the songwriter and provide the reset button to reset their mood and be able to capture again.
- SRS-37: The system shall clear all the input if the songwriter wants to capture the facial expression again.[UI-08-1]
- SRS-38: The system shall analyze the facial expression to determine the Songwriter mood.
- SRS-39: The system shall display “Mood could not be detected, Try again”, if system Unable to determine the Songwriter mood. [UI-08-2]
- SRS-40: The system shall display the detected mood to the Songwriter.
- SRS-41: The system request the song from specific mood playlist in spotify by using spotifyAPI
- SRS-42: The system will store the song by name,album,artist,image,spotify_link and emotion
- SRS-43: The system will match the detected mood with song emotion
- SRS-44: The system shall provide the reset button to reset their mood and be able to capture again.if the songwriter want to reset recommend song and mood detection [UI-08-4]
- SRS-45: The system shall display recommended songs based on the songwriter's detected mood and provide spotify_link ("Listen on spotify"button) in the recommended songs on the playlist.[UI-08-4]

Use Case ID	UC-07		
Use Case name	Capture Facial Expression and Recommended Song		
Created By	NS , WS	Last Update By	NS , WS
Data create	10/05/2024	Last revision	29/05/2024
Actors	Songwriter		
Description	This use case describes the process of capturing facial expressions using the mood detection camera to determine the songwriter's current mood and get the recommended song based on facial expression.		

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Trigger	The songwriter clicks on the "Camera Mood" page		
Preconditions	• Songwriters have access to the website. • Songwriter has granted permission for camera access.		
Use Case Input Specification			
Input	Type	Constraint	Example
Facial expression	Image	Image captured by camera	Facial expression of the image are “Happy” the system will detect the mood “Happy”
Post conditions	Songwriter mood is detected and displayed on the camera view and get recommended songs based on their mood.		
Normal flows	User	System	
	3.Clicks the "capture" button.	1.Display the camera mood page [UI-08] [1A: If the songwriter first time enter to camera mood page] [1B: If the songwriter allow access to camera] [1C: If the songwriter deny access to allow camera] 2.Displays the camera view and provide capture button [UI-08-1] 4.Receive the facial expression of the songwriter and provide the reset button to reset their mood and be able to capture again. [4A: If the songwriter want to capture the facial expression again by clicking reset button]	

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		5.Analyzes the facial expression to determine the Songwriter mood. <i>[5A: Unable to determine the Songwriter mood]</i> 6.Displays the detected mood to the Songwriter [UI-08-3] 7. Request the song from specific mood playlist in spotify by using spotifyAPI 8.Receive specific mood playlist by name,album,artist,image,spotify_link and emotion 9.Store the song by name,album,artist,image,spotify_link and emotion 10.Match the detected mood with song emotion 11.Display recommend songs based on the songwriter's detected mood and provide spotify_link ("Listen on spotify" button) in the recommended songs on the playlist.[UI-08-4] <i>[11A: If songwriter want to reset recommend song and mood detection by clicking reset button]</i>
Alternate Flow	[1A: If the songwriter first time enter to camera mood page] 1. System will display a message to ask permission for the camera access including allow and deny. 2. Go to 1B or 1C [1B: If the songwriter allow access to camera] 1. Go to normal flow 2 [4A: If the songwriter want to capture the facial expression again by clicking reset button]	

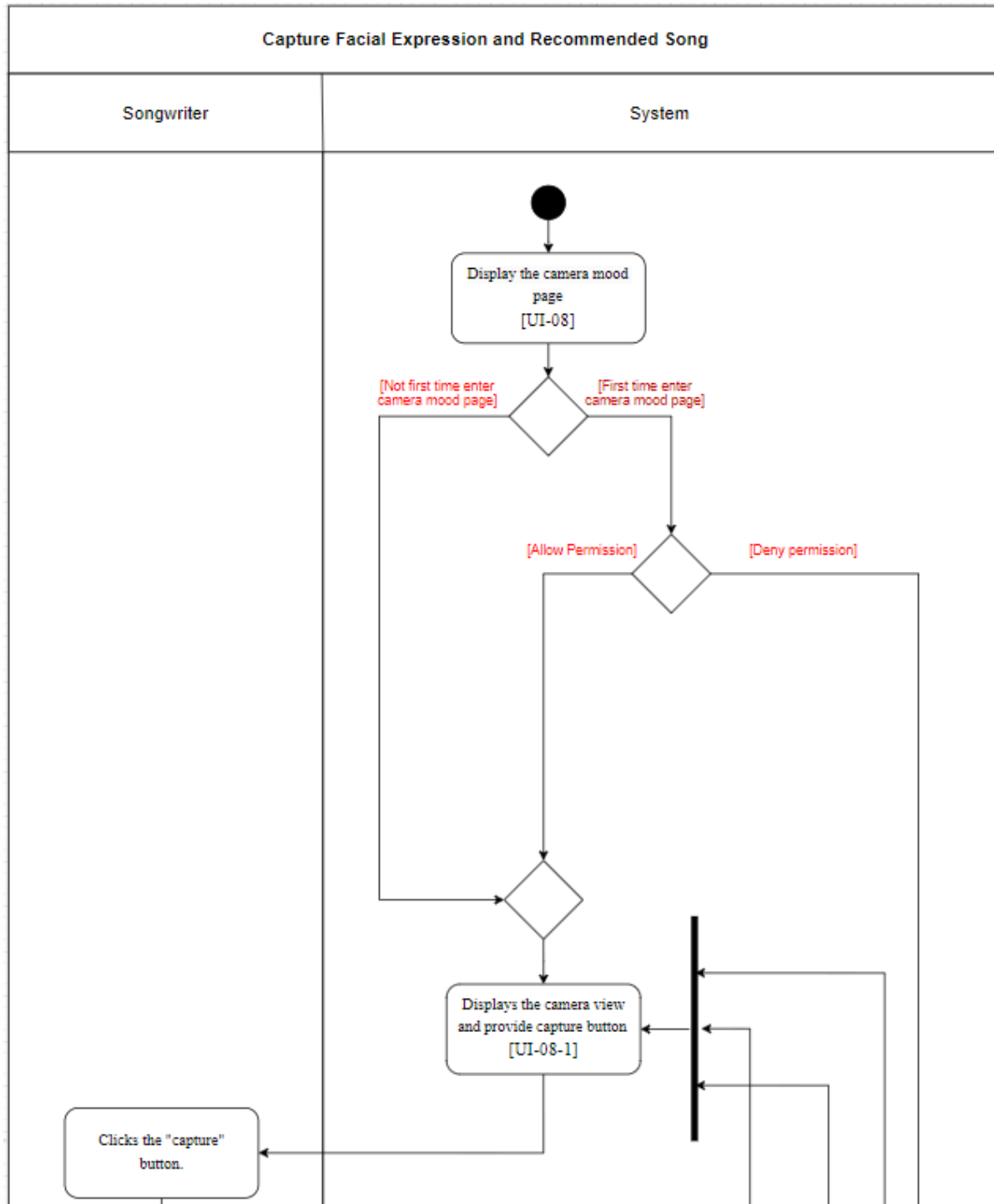
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	1. The songwriter Clicks the "reset" button. 2. The system shall clear all the input. [UI-08-3] 3. Go to normal flow 2 [5A: Unable to determine the Songwriter mood] 1. The system shall display “Mood could not be detected, Try again”, if system unable to determine the Songwriter mood. [UI-08-2] 2. Go to normal flow 2 [11A: If songwriter want to reset recommend song and mood detection by clicking reset button] 1. The songwriter Clicks the "reset" button. 2. The system shall clear all the input if the songwriter wants to capture the facial expression again. [UI-08-1] 3. Go to normal flow 2
Exception Flow	[1C: If the songwriter deny access to allow camera] 1. Flow end in failure case
Assumption	Songwriter device has a functional camera.

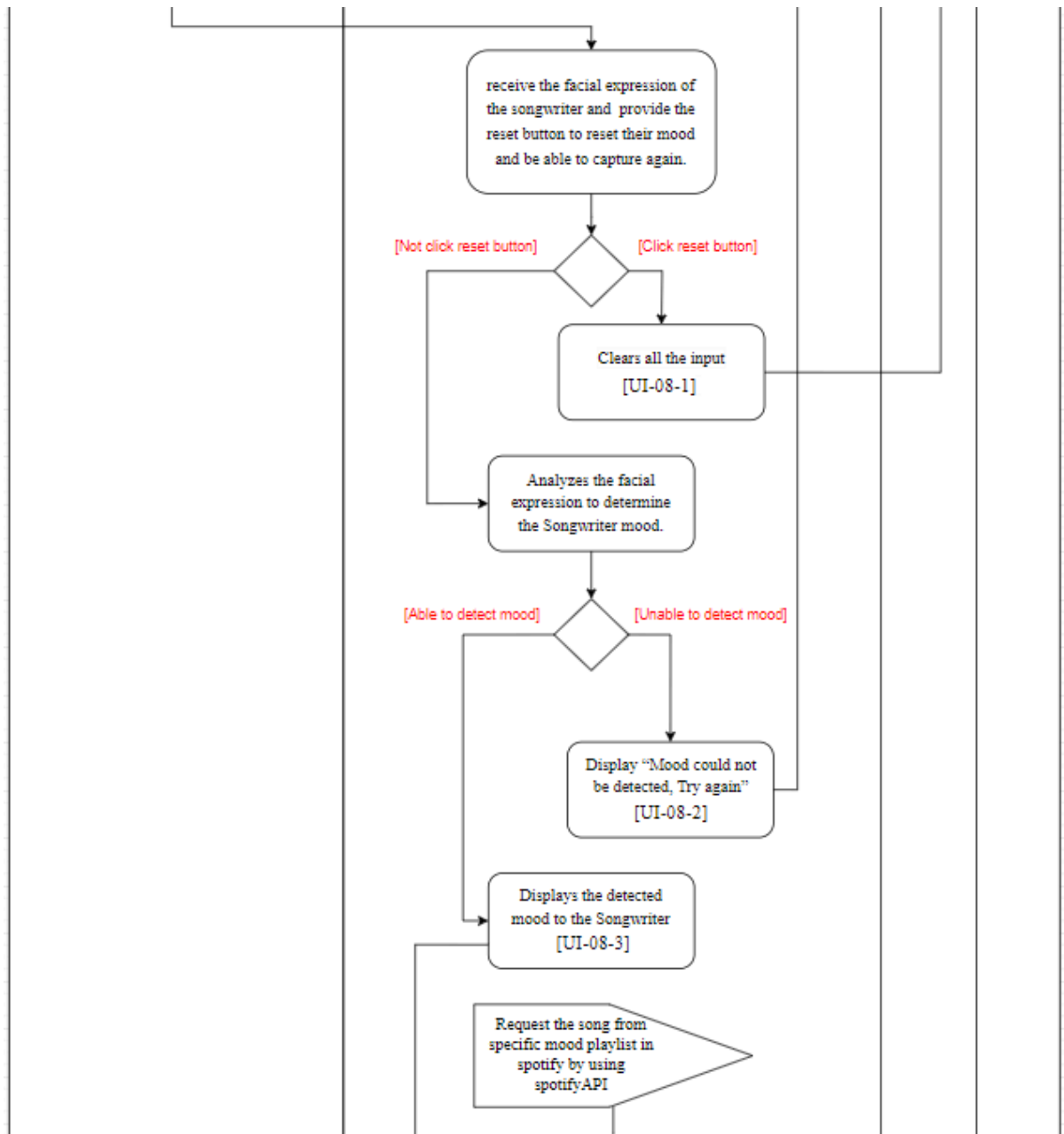
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AD-07: Capture Facial Expression and Recommended Song

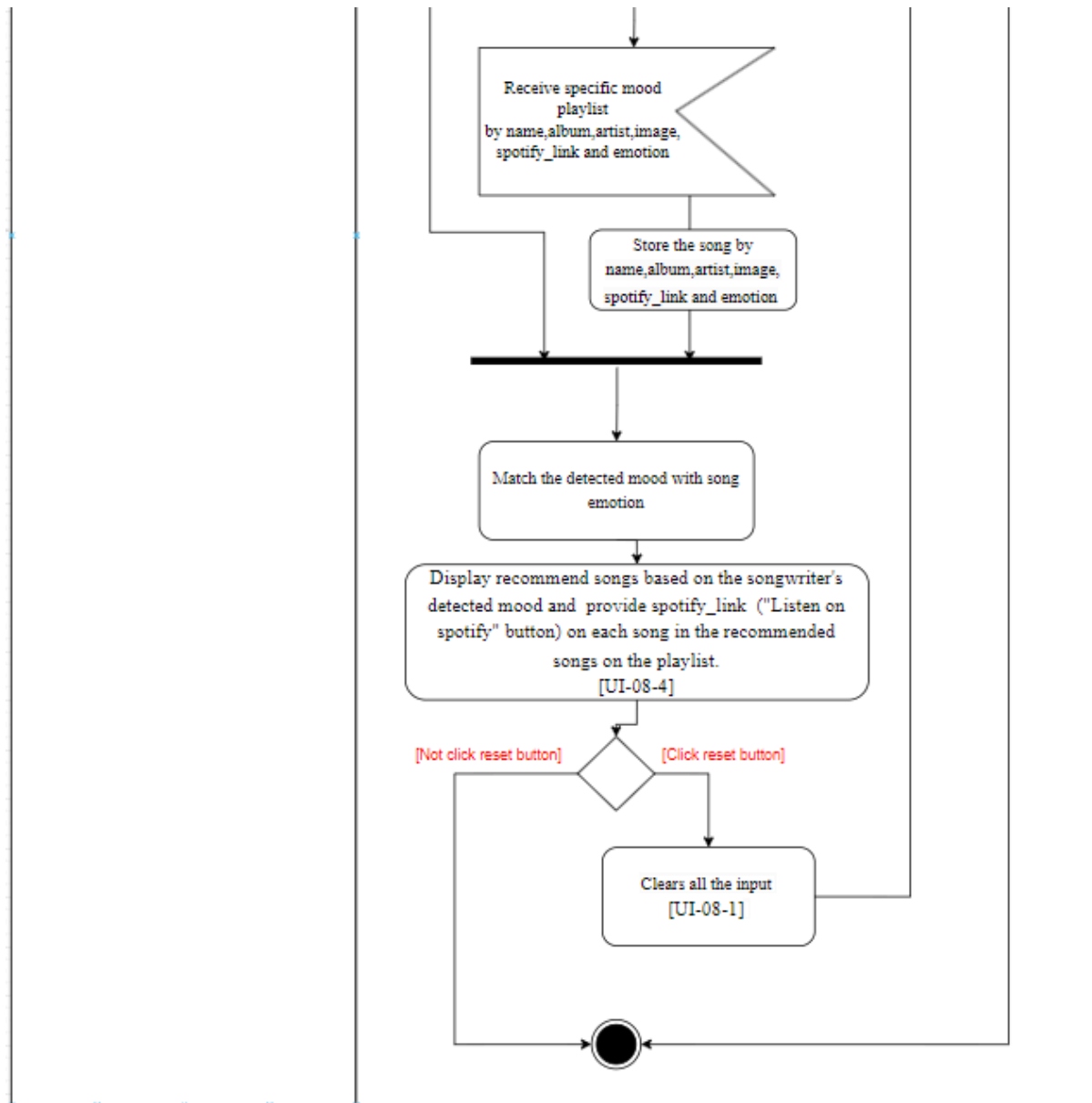
The Capture Facial Expression and Recommended Song activity diagram will show the activity flow of the songwriter having access to the camera and capturing the facial expression into the system to receive the recommended song.



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2.1.3.8 UCD-08

URS-08: Songwriter can select song from recommended song in playlist to listen in spotify

- SRS-46: The system will get spotify_link of each song from database of the song that Songwriter selected
- SRS-47: The system shall redirected Songwriter to Spotify interface by using spotify_link

Use Case ID	UC-08		
Use Case name	Select Song to Listen on Spotify Platform		
Created By	NS , WS	Last Update By	NS , WS
Data create	10/05/2024	Last revision	29/05/2024
Actors	Songwriter		
Description	This use case describes the process of selecting a song from the recommended playlist to listen to on the Spotify platform.		
Trigger	Songwriter Selects a song by clicking spotify_link ("Listen on spotify" button) in the recommended songs on the playlist.		
Preconditions	UC-06		
Use Case Input Specification			
Input	Type	Constraint	Example
Selected song	String	Song selected by the songwriter	"Believer - Imagine dragon"
Post conditions	● Songwriters are redirected to the Spotify application to listen to their selected song.		

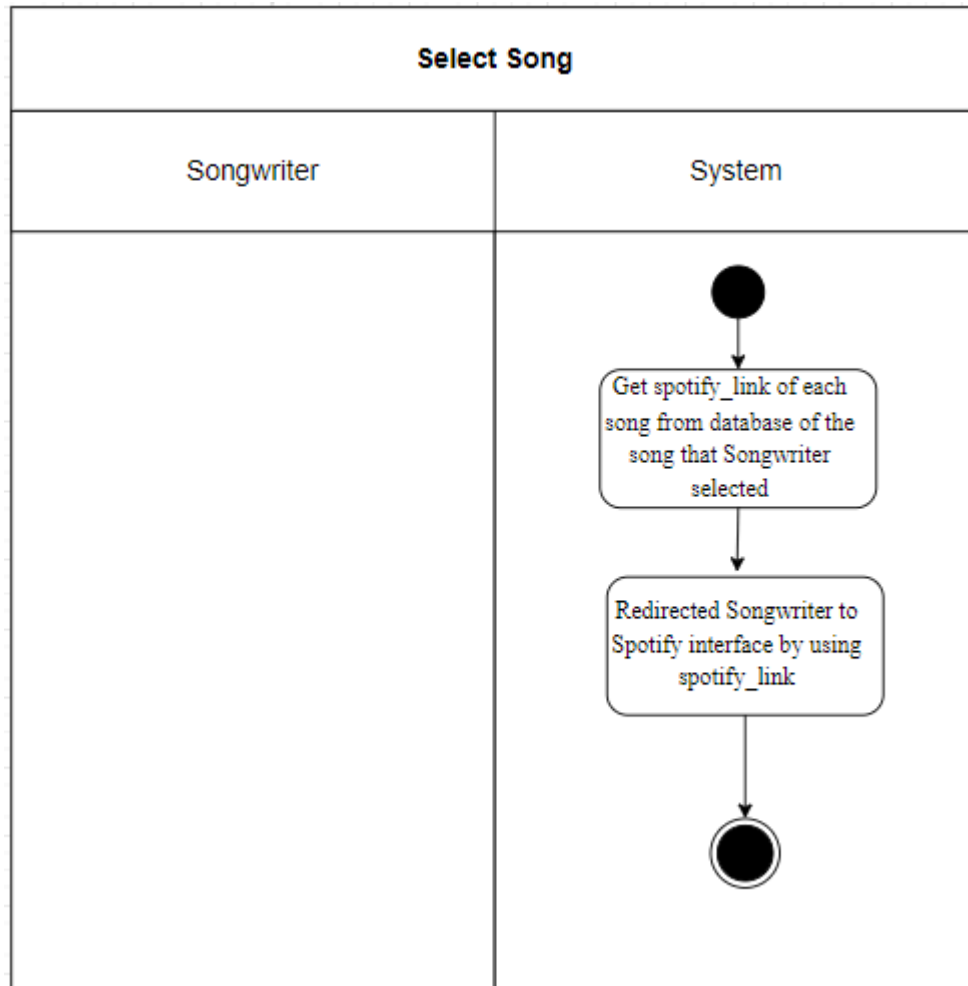
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Normal flows	User	System
		1.Get spotify_link of each song from database of the song that Songwriter selected 2.Redirected Songwriter to Spotify interface by using spotify_link
Alternate Flow	-	
Exception Flow	-	
Assumption	The songwriter has an internet connection to listen their song on spotify platform	

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AD-08: Select Song

The Select Song activity diagram will show the activity flow of the songwriter received the recommended song based on songwriter mood and can redirected to the spotify platform



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2.2.3 Use Case Description and Activity Diagram

2.2.3.09 UCD-09 : Access and View Chord Progression Information

URS-09: The songwriter can access the "Chord Progression" page to view information, selecting from the navigation options provided in SRS-02.

- SRS-48: The system shall display the "Chord Progression" page [UI-09] that includes introductory information about "What is a Chord Progression?" and lists available chord progressions (2-5-1-6, 1-6-2-5, 1-5-6-4, 1-6-4-5, 1-4-5-1, 1-4-6-5, 1-6-2-6, 6-4-3-2).
- SRS-49: The system shall provide a "Back" button [UI-09] on the detailed chord progression information page to allow the songwriter to return to the previous page or the main "Chord Progression" page.
- SRS-50: The system shall provide a "Next" button [UI-09] on the detailed chord progression information page to allow the songwriter to navigate between different chord progression details [UI-09-1].

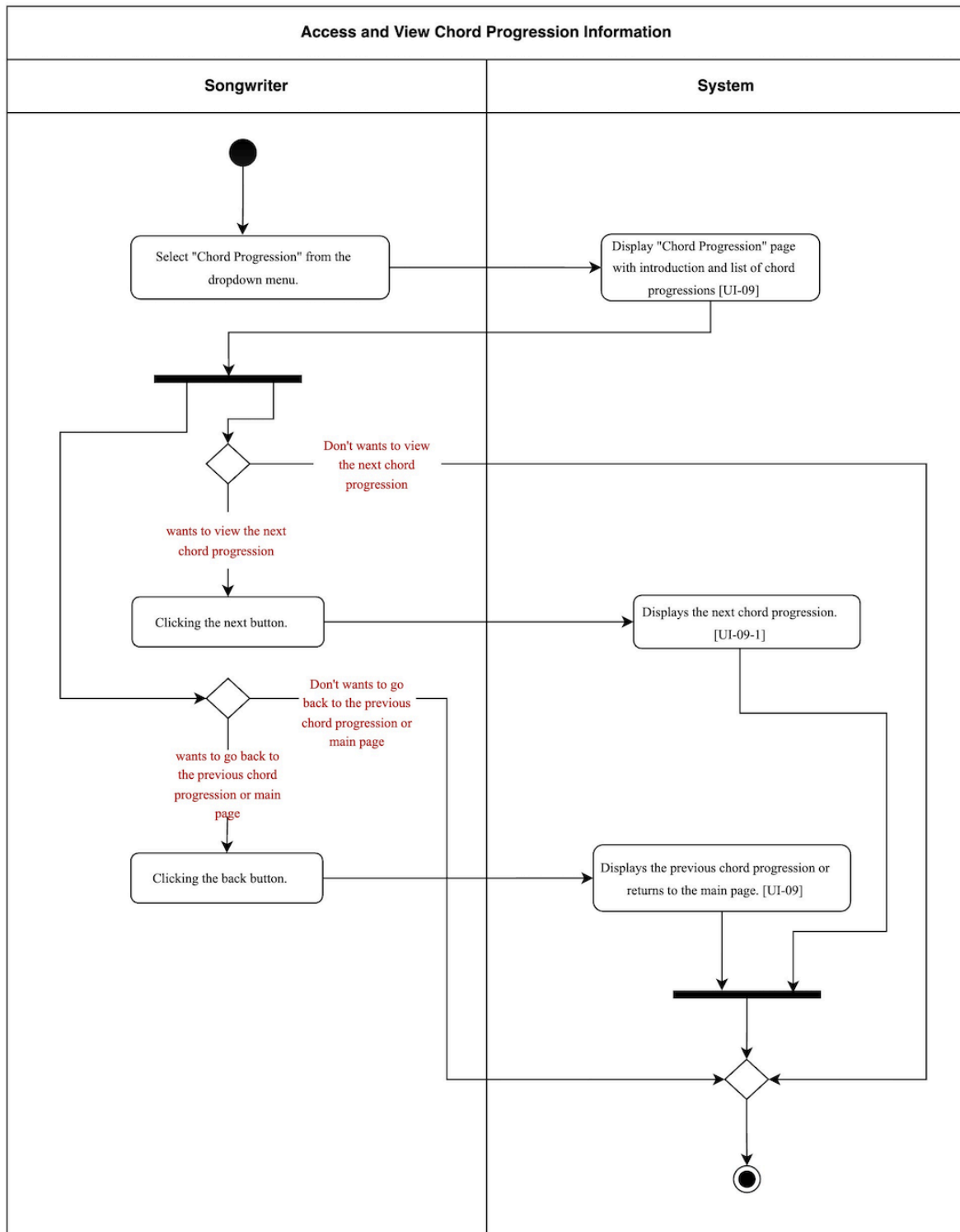
Use Case ID	UC-09		
Use Case name	Access and View Chord Progression Information		
Created By	NS, WS	Last Update By	NS, WS
Data create	10/08/2024	Last revision	30/08/2024
Actors	Songwriter		
Description	This use case describes how the songwriter accesses the "Chord Progression" page to view detailed information about chord progressions, including an introduction and a list of available progressions.		
Trigger	The songwriter selects the "Chord Progression" option from the drop-down menu in the navigation bar, as provided in SRS-02. [UI-02]		
Preconditions	The system has loaded the "navbar-toggler-icon" and the drop-down menu as described in SRS-02. [UI-02]		
Use Case Input Specification			
Input	Type	Constraint	Example
-	-	-	-

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Post conditions	The songwriter is directed to the "Chord Progression" page displaying introductory information and a list of available chord progressions. [UI-09]	
Normal flows	User	System
	1. Select "Chord Progression" from the dropdown menu.	2. Display the "Chord Progression" page that includes introductory information about "What is a Chord Progression?" and lists available chord progressions (2-5-1-6,1-6-2-5,1-5-6-4,1-6-4-5,1-4-5-1,1-4-6-5, 1-6-2-6, 6-4-3-2). [UI-09] (5A: If the songwriter wants to view the next chord progression by clicking the next button.) (5B: If the songwriter wants to go back to the previous chord progression or main page by clicking the back button.)
Alternate Flow	[5A: If the songwriter wants to view the next chord progression] - The songwriter Clicks the next button. - Displays the next chord progression. [UI-09-1] - Go to normal flow 5 [5B: If the songwriter wants to go back to the previous chord progression or main page by clicking the back button] - The songwriter Clicks the back button. - Displays the previous chord progression or returns to the main page. [UI-09] - Go to normal flow 5	
Exception Flow	-	
Assumption	The songwriter has an internet connection	

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AD-09: Access and View Chord Progression Information



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2.2.3.10 UCD-10: View and interact the chord progression chart

URS-10: The songwriter can view the chord progression chart after accessing the "Chord Progression" page.

- SRS-51: The system shall display a "CLICK HERE TO SEE CHORD PROGRESSION CHART" button on UI-09 when the songwriter accesses the "Chord Progression" page. [UI-09]
- SRS-52: After the "CLICK HERE TO SEE CHORD PROGRESSION CHART" button is clicked on UI-09, the system shall navigate to a new page that displays a chart showing various chord progressions (2-5-1-6, 1-6-2-5, 1-5-6-4, 1-6-4-5, 1-4-5-1, 1-4-6-5, 1-6-2-6, 6-4-3-2) and their keys (C, D, E, F, G, A, B). [UI-10]
- SRS-53: The system shall allow the songwriter to select any progression from the table. After selection, the system shall highlight the selected progression and key in yellow. [UI-10-1]
- SRS-54: After a chord progression is selected, the system shall display a series chord of guitar diagrams corresponding to the selected progression below the table. [UI-10-1]
- SRS-55: The system shall provide sound [UI-10-1] for the songwriter to listen to the piano audio playback of the selected chord progression if the "sound" button is clicked. [UI-10-2]
- SRS-56: The system shall allow the songwriter to change their selection chord of guitar, updating the highlighted progression and the displayed chord diagrams accordingly.

Use Case ID	UC-10		
Use Case name	View and interact the chord progression chart		
Created By	NS, WS	Last Update By	NS, WS
Data create	10/08/2024	Last revision	30/08/2024
Actors	Songwriter		
Description	This use case describes how a songwriter interacts with the "Chord Progression" page. The songwriter can view, select, and listen to various chord progressions within a chart.		
Trigger	The songwriter accesses the "Chord Progression" page and clicks the "CLICK HERE TO SEE CHORD PROGRESSION CHART" button. [UI-09]		
Preconditions	UC-09		
Use Case Input Specification			
Input	Type	Constraint	Example

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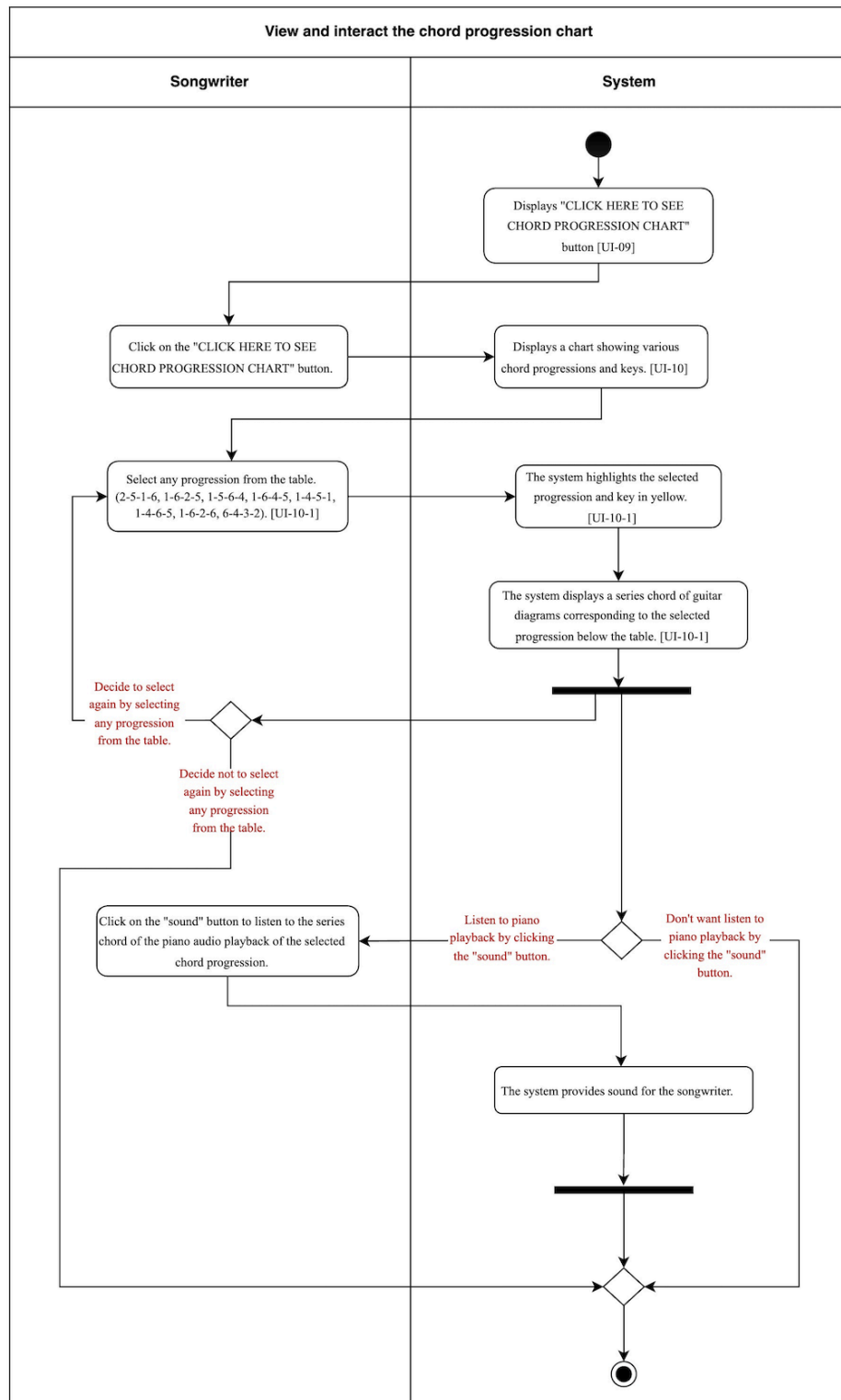
-	-	-	-
Post conditions	The selected chord progression is highlighted in yellow, with corresponding chord diagrams displayed below the chart and audio playback provided for the selected progression. [UI-10-1]		
Normal flows	User	System	
	2. Click on the "CLICK HERE TO SEE CHORD PROGRESSION CHART" button. 4. Select any progression from the table. (2-5-1-6, 1-6-2-5, 1-5-6-4, 1-6-4-5, 1-4-5-1, 1-4-6-5, 1-6-2-6, 6-4-3-2).	1. The system displays "CLICK HERE TO SEE CHORD PROGRESSION CHART" button [UI-09] 3. The system displays a chart showing various chord progressions and keys. [UI-10] 5. The system highlights the selected progression and key in yellow. [UI-10-1] 6. The system displays a series chord of guitar diagrams corresponding to the selected progression below the table. [UI-10-1] <i>(6A: If the songwriter wants to select by selecting any progression from the table.)</i> <i>(6B: If the songwriter wants to listen to the piano audio playback of the selected chord progression by clicking the "sound" button)</i>	
Alternate Flow	[6A: If the songwriter wants to select by selecting any progression from the table.] 1. Go to normal flow 4. [6B: If the songwriter wants to listen to the piano audio playback of the selected chord progression by clicking the "sound" button]		

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	1. The system provides sound for the piano audio playback of the selected chord progression. 2. The use case ends.
Exception Flow	-
Assumption	The songwriter has an internet connection

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AD-10: View and interact the chord progression chart



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2.2.3.11 UCD-11: Input Lyrics and View Musical Analysis

URS-11: The songwriter can input lyrics into a text field and view a detailed musical analysis based on the lyrics input, selecting from the navigation options provided in SRS-02.

- SRS-57: The system shall display a "AI Analysis" page [UI-11]
- SRS-58: The system shall provide a text field for the songwriter to input lyrics.
- SRS-59: The system shall analyze the input lyrics and chords, tempo, time signature, keys, and extracts musical information based on the provided lyrics. [UI-11-2]
- SRS-60: If the songwriter attempts to proceed without completing the input lyrics fields, the system shall display an error message indicating the missing fields. The error message will state: "The lyrics field cannot be empty. Enter your lyrics to proceed." [UI-11-1]
- SRS-61: If there is no internet connection, the system shall display an error message "Internet no connection" [UI-11-3]
- SRS-62: The system shall allow the songwriter to analyze new lyrics by filling in the text field again on the AI Analysis page.
- SRS-63: The system shall display the results of the analysis to the songwriter based on the input lyrics. [UI-11-5]

Use Case ID	UC-13		
Use Case name	Input Lyrics and View Musical Analysis		
Created By	NS, WS	Last Update By	NS, WS
Data create	10/08/2024	Last revision	30/08/2024
Actors	Songwriter		
Description	This use case describes the process through which the songwriter can input lyrics into a text field and receive a detailed musical analysis based on the input lyrics.		
Trigger	The songwriter clicks on the "AI Analysis" page. [UI-02]		
Preconditions	The system has loaded the "navbar-toggler-icon" and the drop-down menu as described in SRS-02. [UI-02]		
Use Case Input Specification			
Input	Type	Constraint	Example
Lyrics	Text	Required	In the quiet night, under the moon's soft glow,

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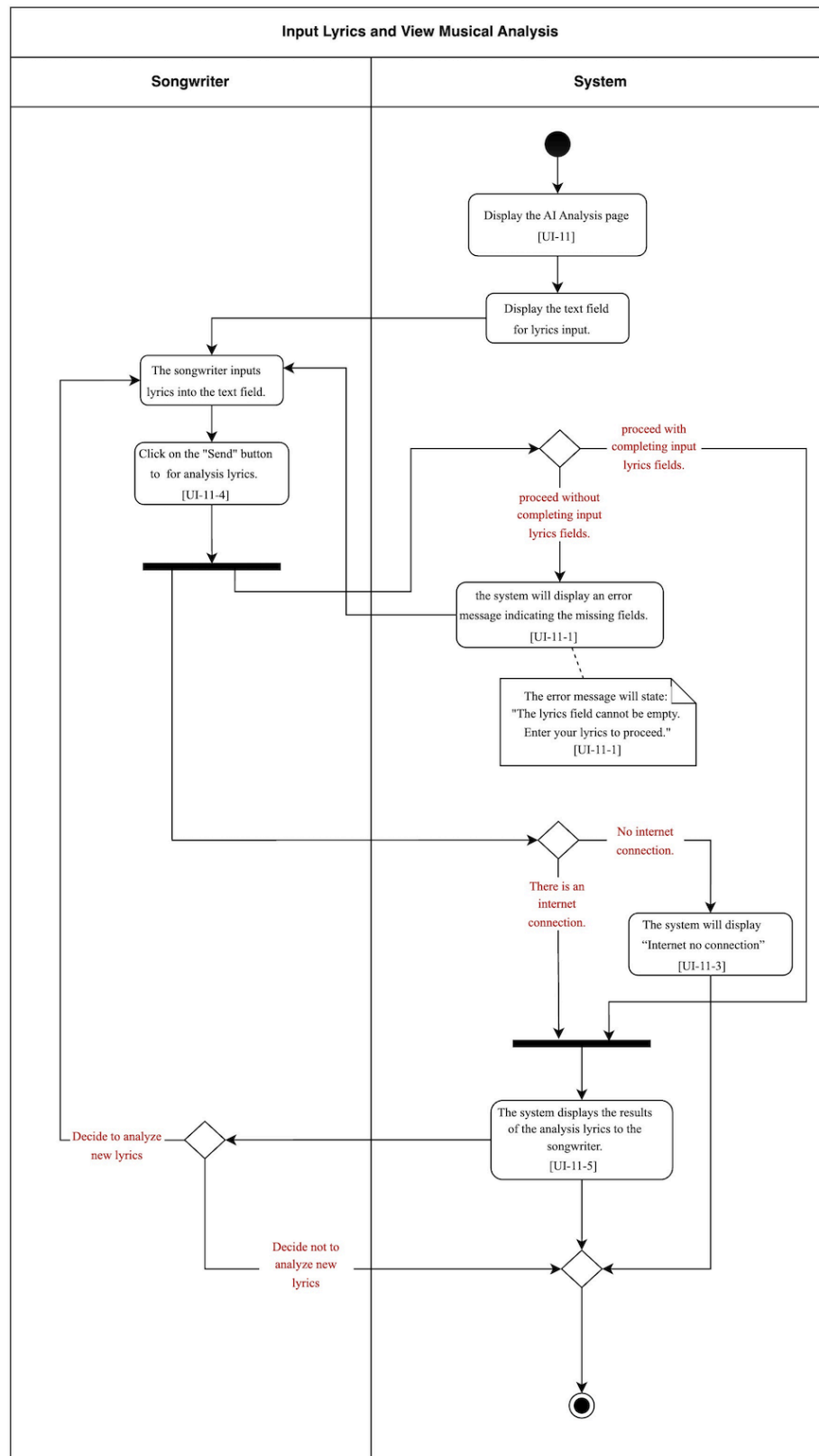
			We dance through the dark, where dreams freely flow. Your hand in mine, we'll reach for the skies, With every heartbeat, our love never dies. Through the shadows and light, we'll find our way, In the whispers of dawn, where night turns to day. Together we'll soar, on wings made of dreams, In this endless journey, nothing's as it seems. Every star that we chase, will guide us through, In the canvas of night, painted just for you. Hold on to me tight, let the magic unfold, In this story of love, we're forever bold.
Post conditions	The songwriter receives a detailed musical analysis including suggested chords, recommended tempo, time signature, key identification, and other relevant musical information. [UI-11-5]		
Normal flows	User	System	

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	3.The songwriter inputs lyrics into the text field. 4.Click on the "Send" button to for analysis lyrics. [UI-11-4]	1.Display the AI Analysis page 2.Display the text field for lyrics input. 5. Check if all required input lyrics are completed and meet the constraints. <i>(5A: If the songwriter attempts to proceed without completing input lyrics fields.)</i> 6.The system displays the results of the analysis lyrics to the songwriter. <i>(6A: Unable to displays the results of the analysis lyrics causing by no internet connection)</i> <i>(6B: If the songwriter wants to analyze new lyrics by filling lyrics in the text field.)</i>
Alternate Flow	[5A: If the songwriter attempts to proceed without completing input lyrics fields.] - the system will display an error message indicating the missing fields.The error message will state: "The lyrics field cannot be empty. Enter your lyrics to proceed." [UI-11-1] - Go to normal flow 3. [6B: If the songwriter wants to analyze new lyrics by filling lyrics in the text field.] - Go to normal flow 3.	
Exception Flow	[6A: Unable to displays the results of the analysis lyrics causing by no internet connection] - The system will display “Internet no connection” [UI-11-3] - The use case ends.	
Assumption	The songwriter has an internet connection	

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AD-11: Input Lyrics and View Musical Analysis



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2.2.3.12 UCD-12: Save Analyzed Lyrics

URS-12: The songwriter can save information as a .txt file on their local computer after analyzing the input lyrics.

- SRS-64: The system shall provide a "Save" button that allows the songwriter to save the analyzed lyrics information.
- SRS-65: The system shall save the analyzed lyrics information to the songwriter's local computer in a .txt file format. [UI-11-6]
- SRS-66: If there is no internet connection, the system shall display an error message "No internet connection" [UI-11-7]

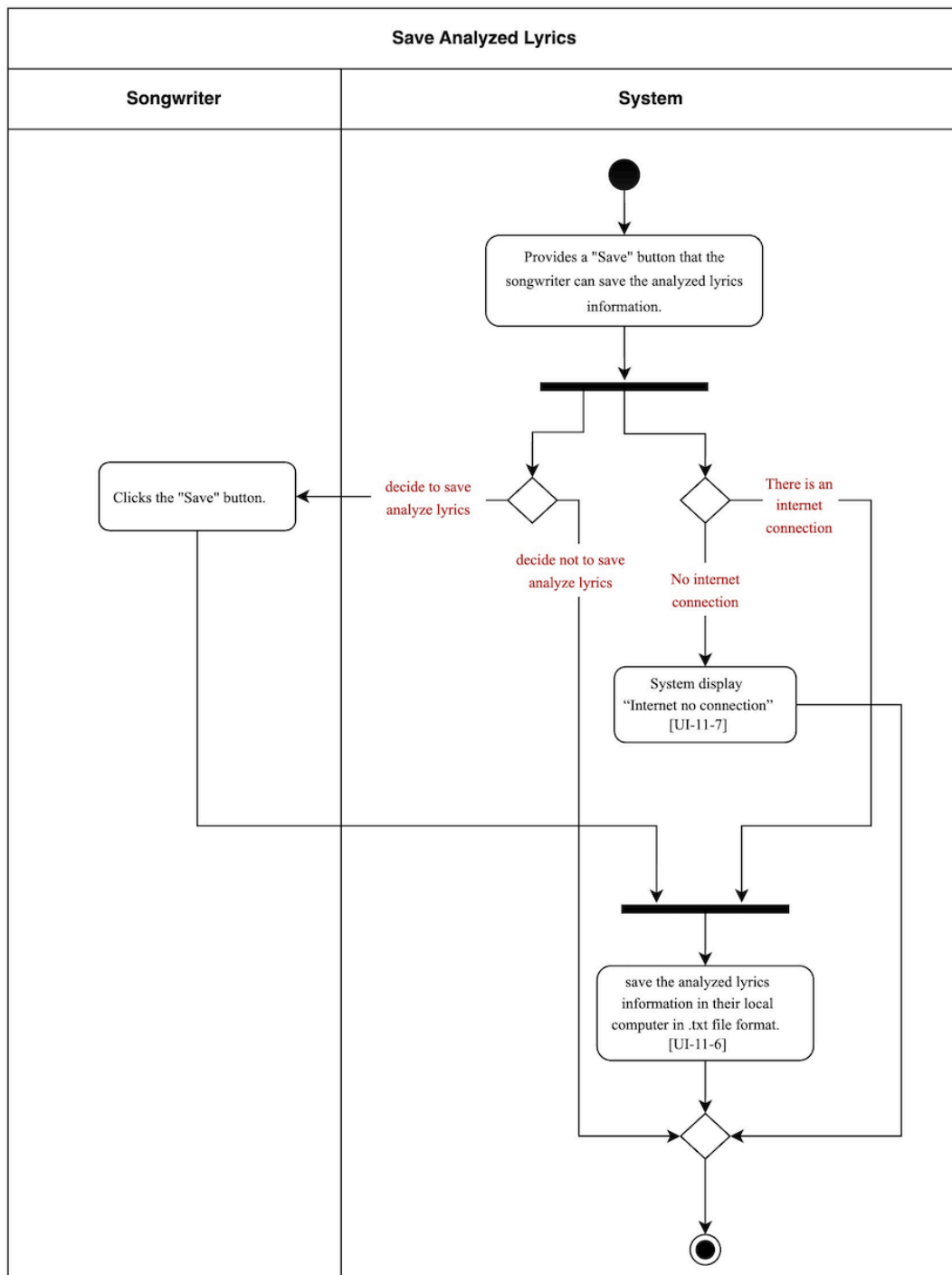
Use Case ID	UC-14		
Use Case name	Save Analyzed Lyrics		
Created By	NS, WS	Last Update By	NS, WS
Data create	10/08/2024	Last revision	30/08/2024
Actors	Songwriter		
Description	This use case describes the process through which the songwriter can save the analyzed lyrics information as a .txt file on their local computer.		
Trigger	The songwriter clicks the "Save" button after the lyrics analysis has been displayed.		
Preconditions	UC-11		
Use Case Input Specification			
Input	Type	Constraint	Example
Save action	button	User-initiated	Clicking "Save"
Post conditions	The analyzed lyrics information are saved as a .txt file on the songwriter's local computer. [UI-11-6]		
Normal flows	User	System	

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	2.Clicks the "Save" button.	1.Provides a "Save" button that the songwriter can save the analyzed lyrics information. 3.Saves the analyzed lyrics information on their local computer in .txt file. <i>(3A: Unable to save the analyzed lyrics information causing by no internet connection)</i>
Alternate Flow	-	
Exception Flow	[3A: Unable to save the analyzed lyrics information causing by no internet connection] 1. The system will display “No internet connection” [UI-11-7] 2. The use case ends.	
Assumption	The songwriter has an internet connection	

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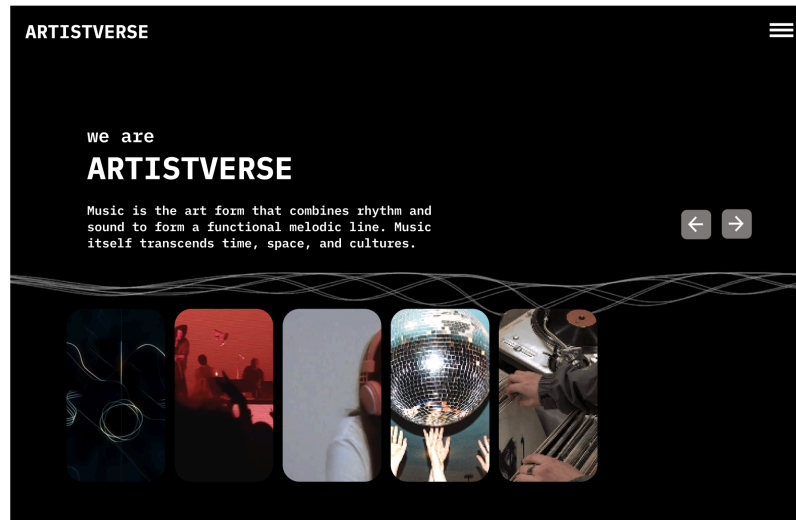
AD-12: Save Analyzed Lyrics



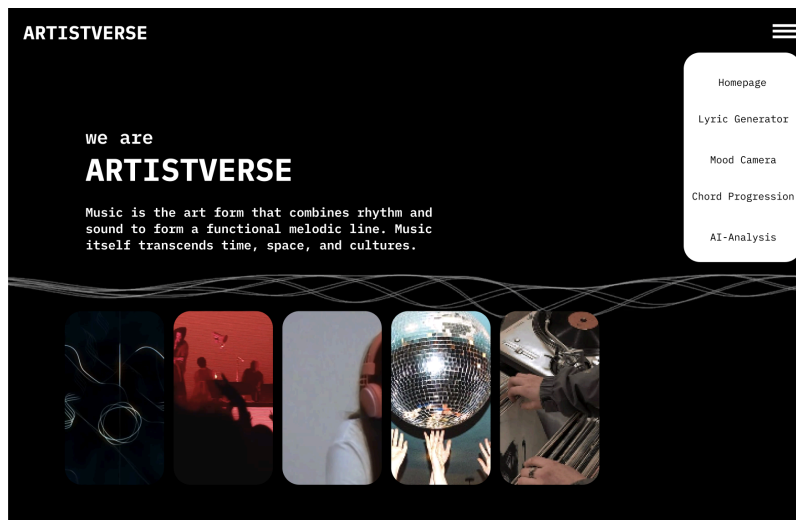
Document Name	Artistverse-SRS V.3.0	Owner	NS , WS	Page	64
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APPENDIX UI

UI-01 Home page



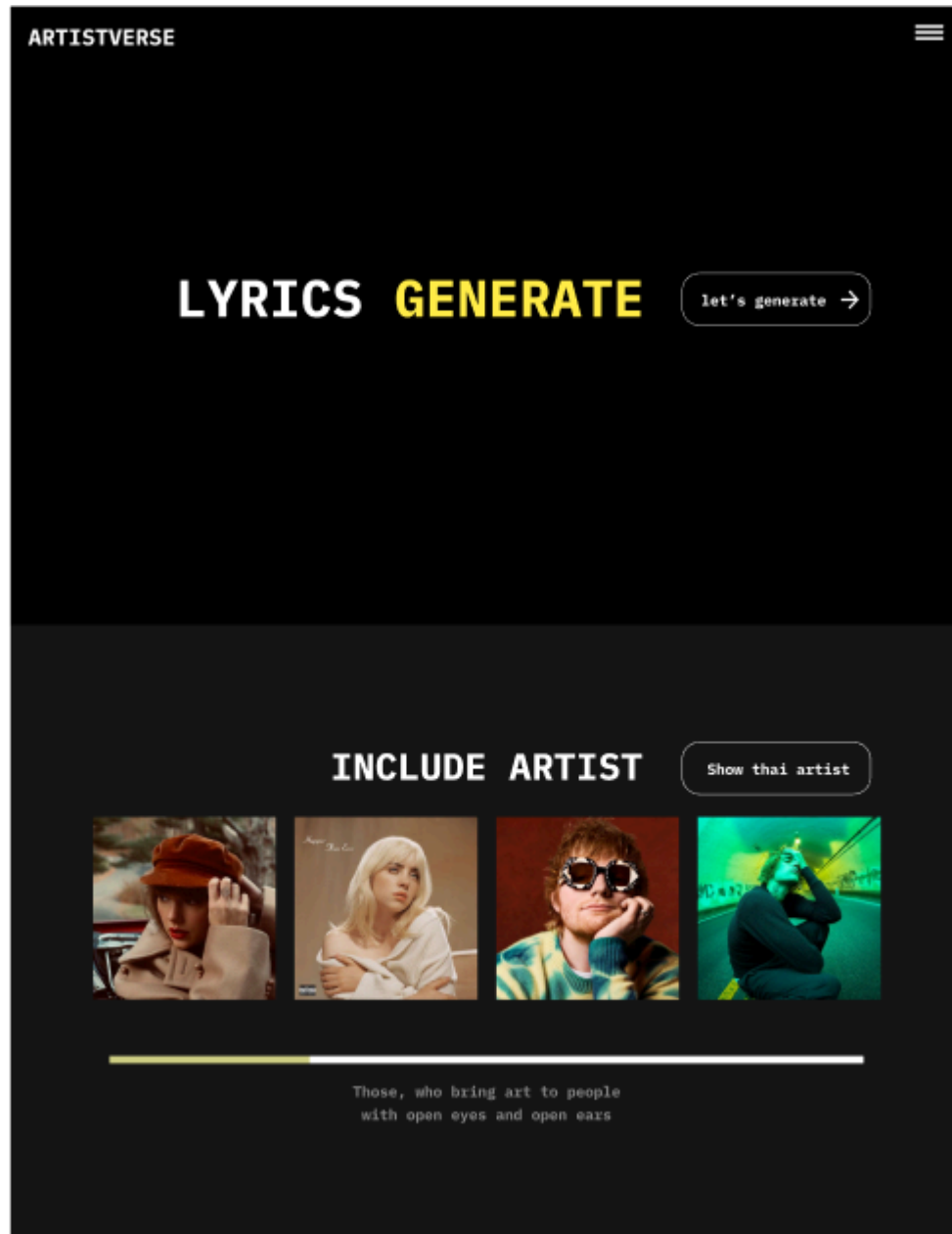
UI-02 Home page with "navbar-toggler-icon" button



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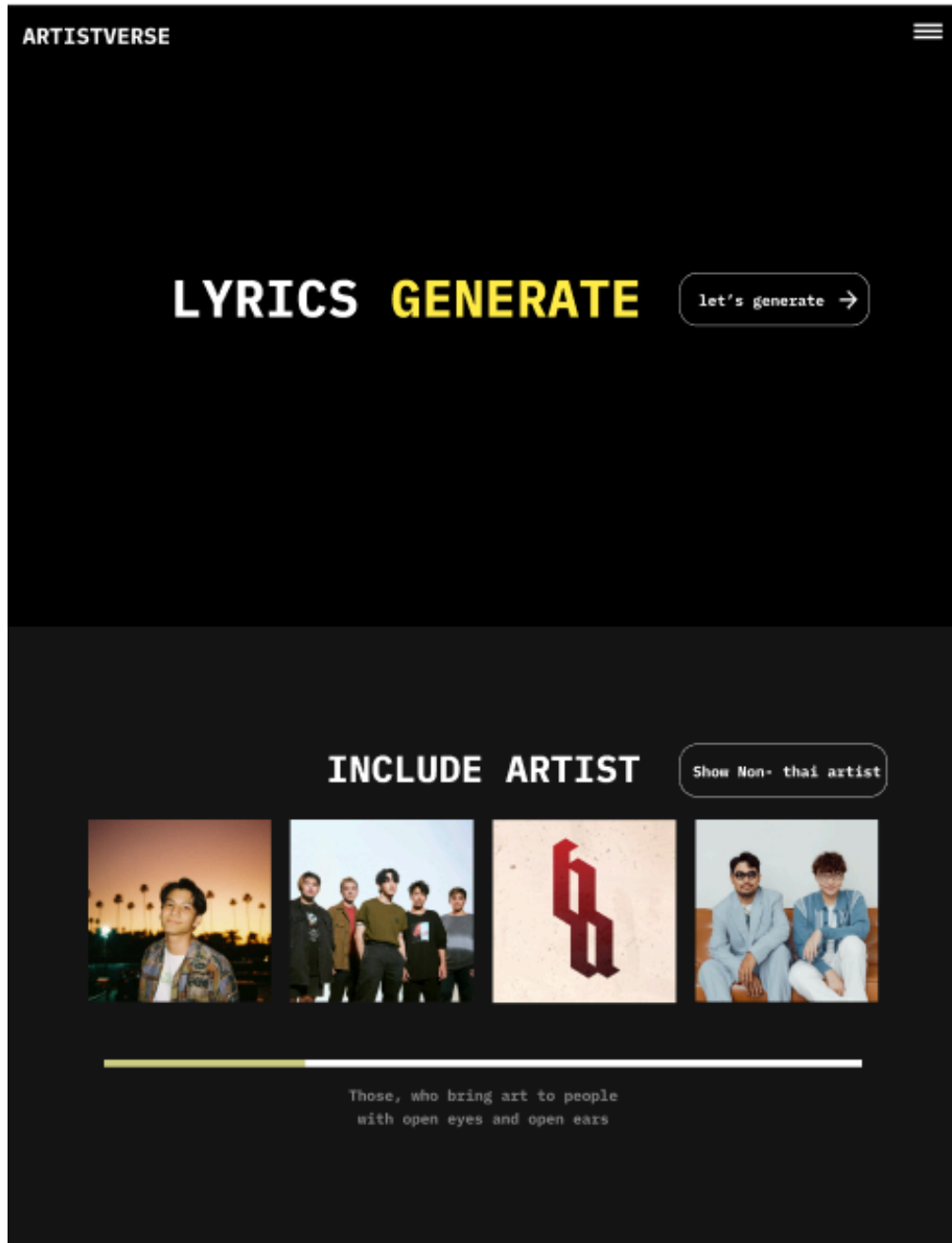
UI-03 Lyric Generator page with "Let's generate" button that can redirect the Songwriter to the lyrics generator by Genre page and information about including the Non-thai artist.

X



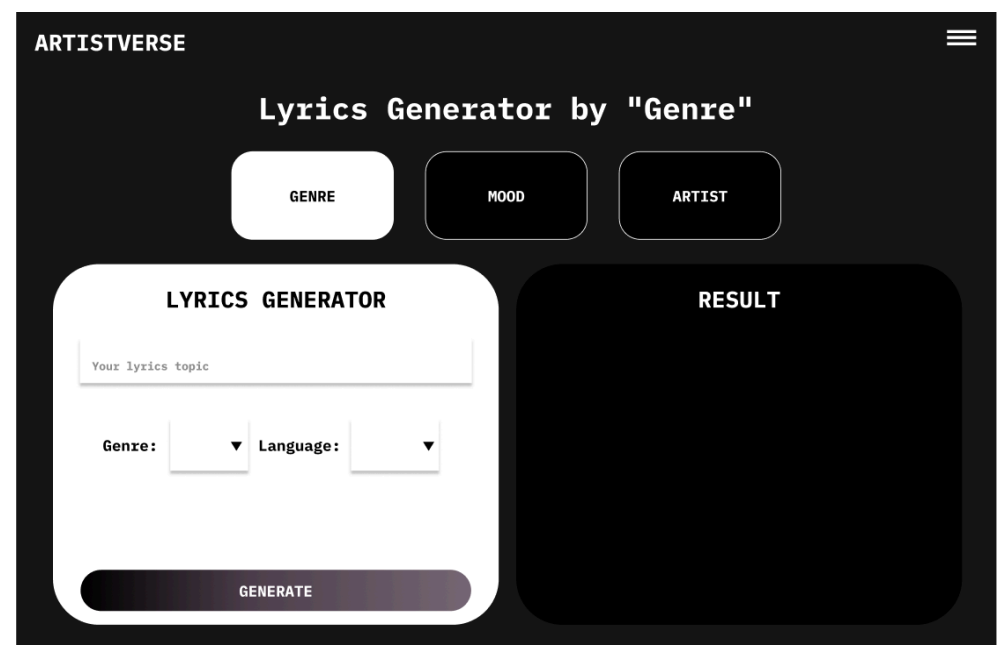
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UI-03-1 Lyric Generator page with "Let's generate" button that can redirect the Songwriter to the lyrics generator by Genre page and information about including the Thai artist.

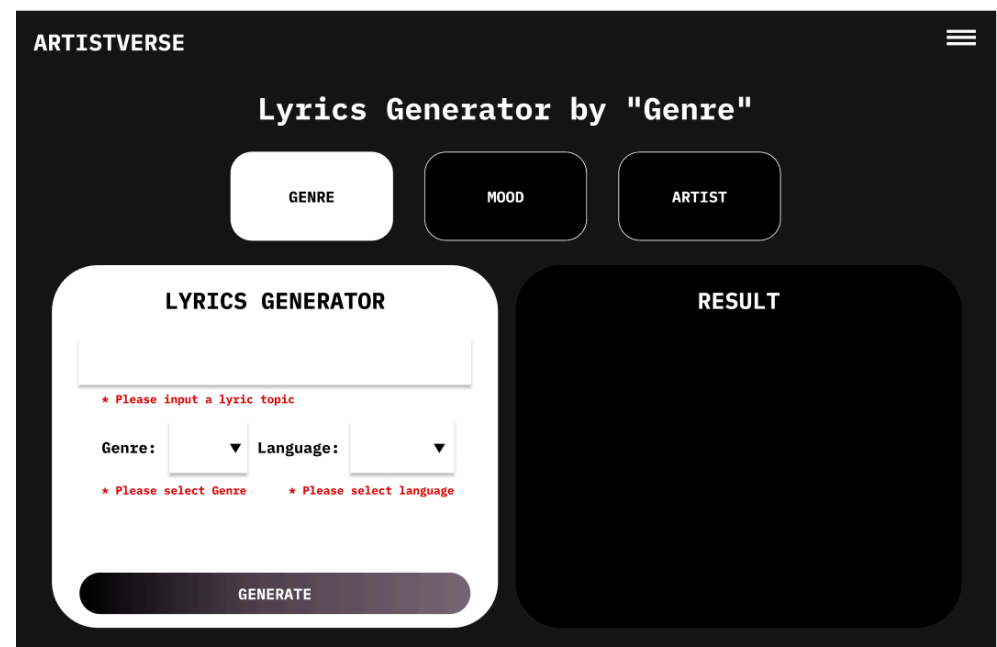


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UI-04 Lyric Generator by Genre Page

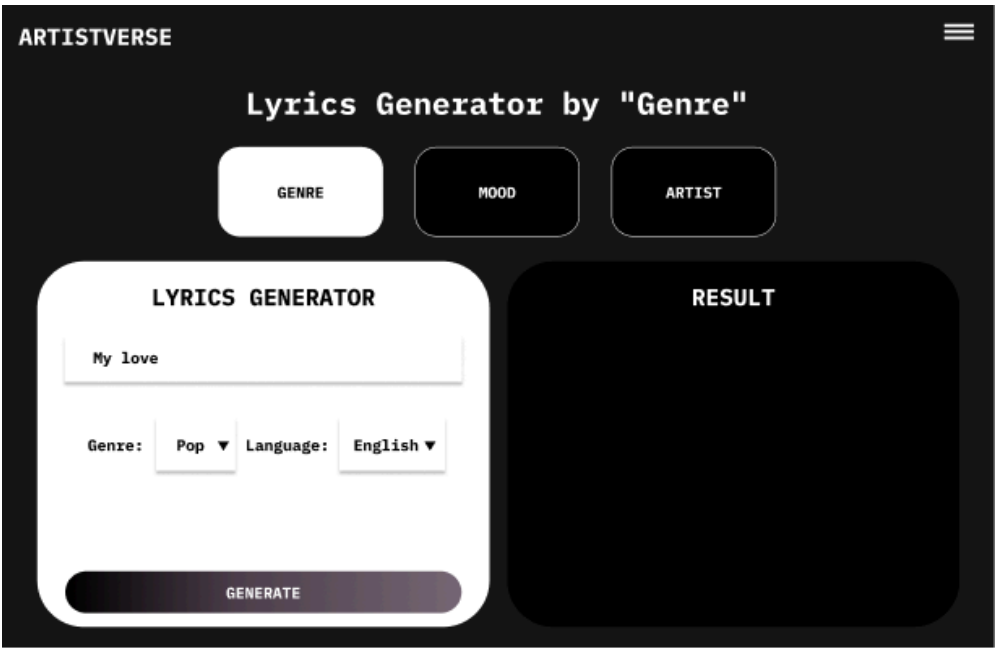


UI-04-1 With invalid input

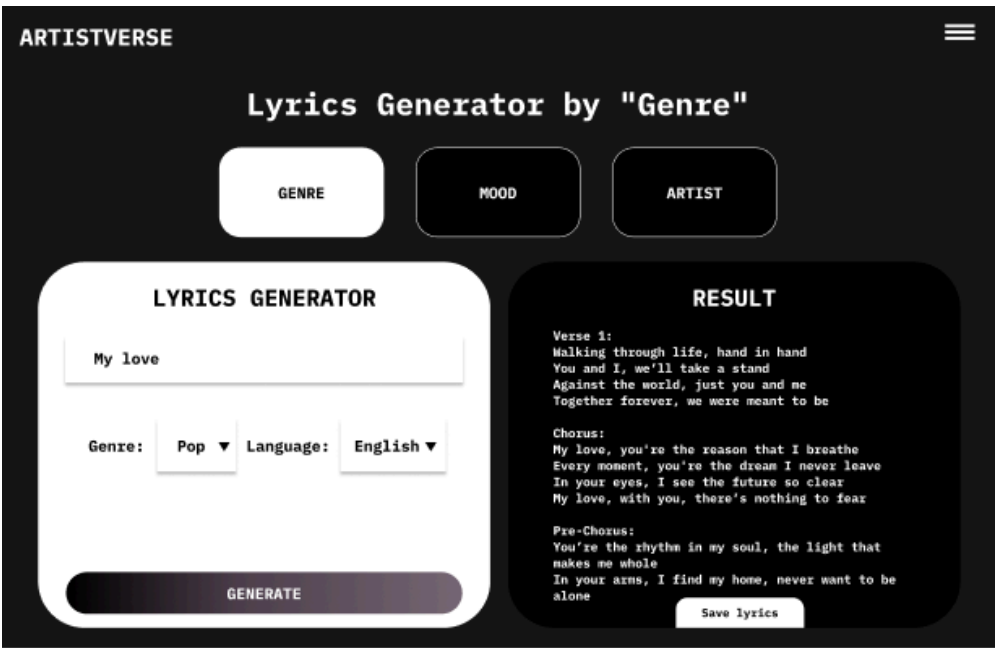


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UI-04-2 With input information

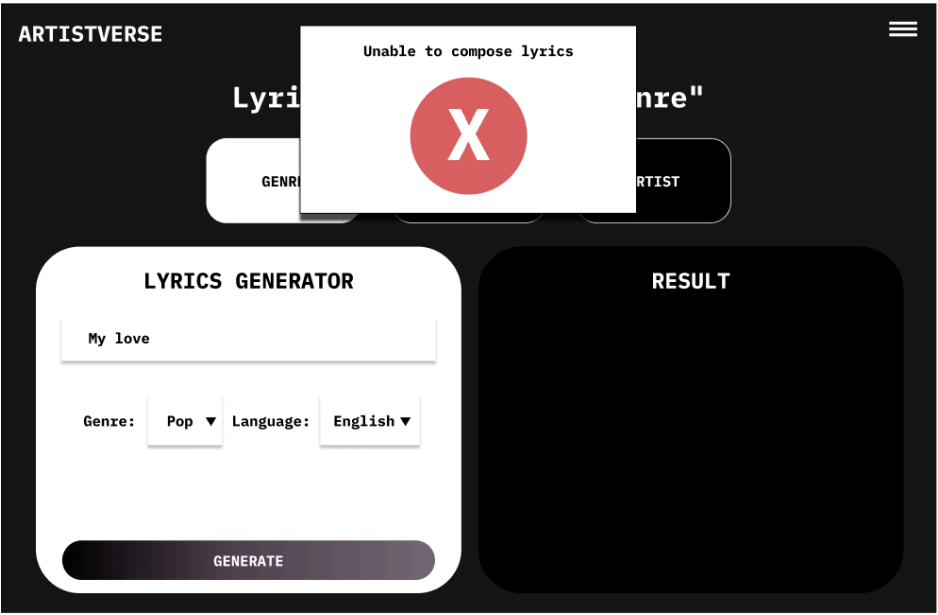


UI-04-3 With display the created lyrics

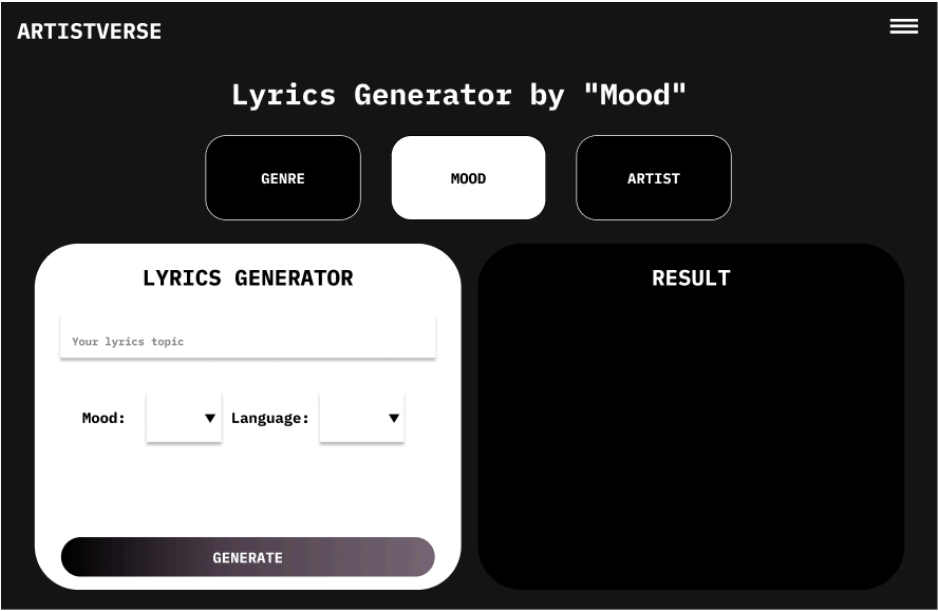


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UI-04-4 With causing by no internet connection

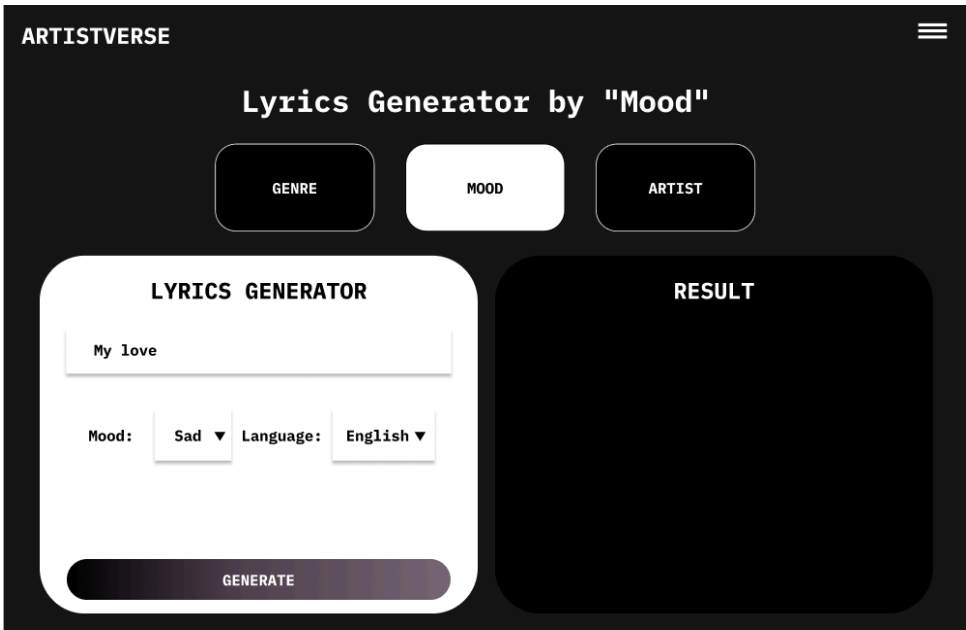


UI-05 Lyric Generator by Mood page

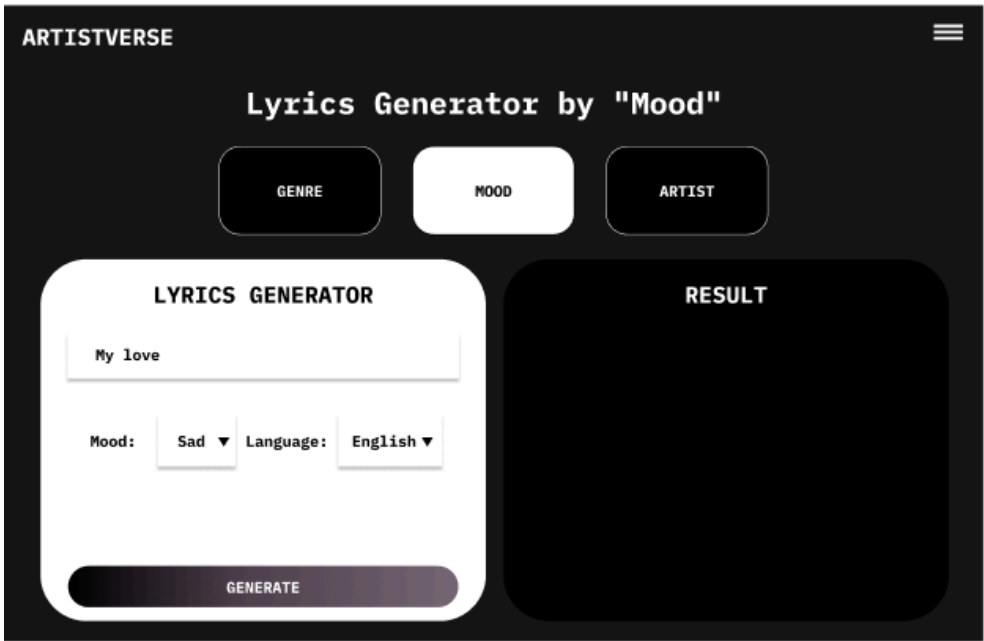


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UI-05-1 With input information

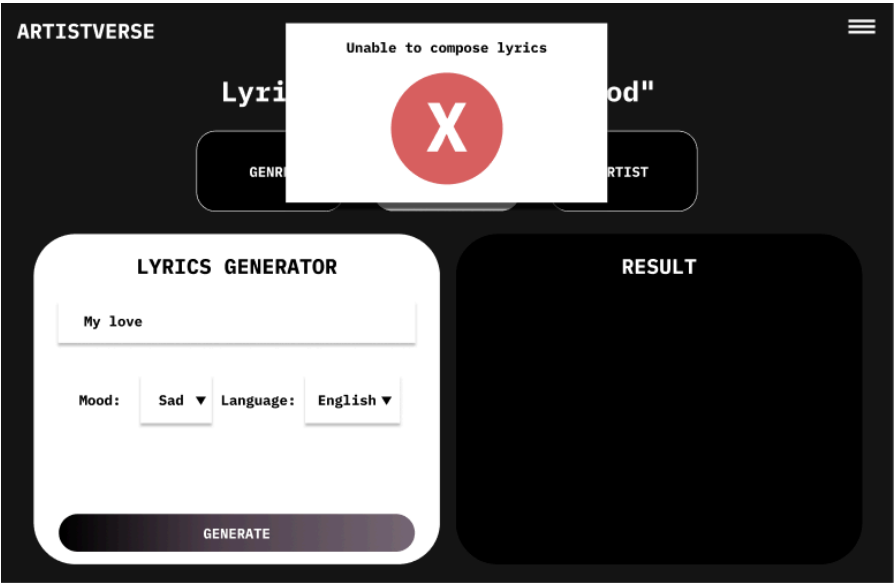


UI-05-2 With invalid input

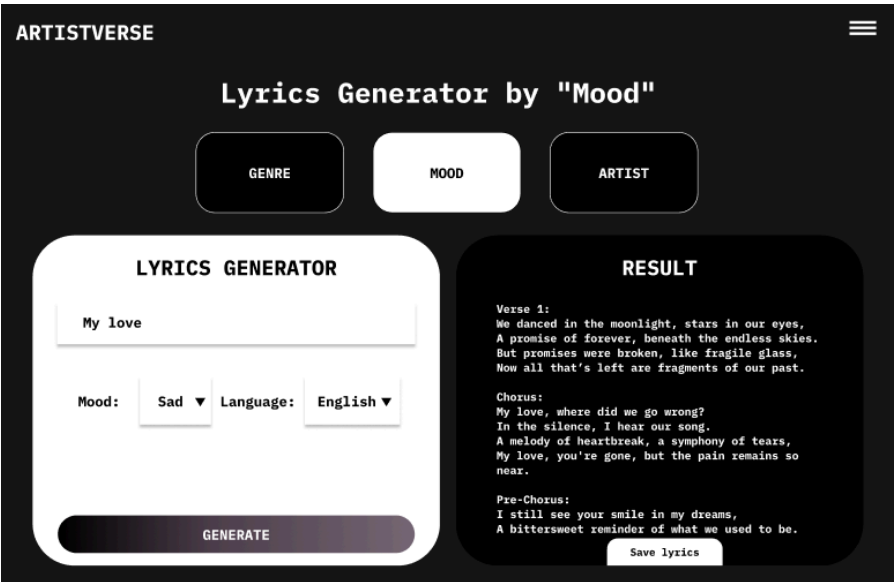


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UI-05-3 With causing by no internet connection



UI-05-4 With display the created lyrics



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UI-06 Lyric Generator by Style Artist Page

ARTISTVERSE

Lyrics Generator by "Style Artist"

GENRE

MOOD

ARTIST

LYRICS GENERATOR

Your lyrics topic

Artist:

Language:

GENERATE

RESULT

UI-06-1 With invalid input

ARTISTVERSE

Lyrics Generator by "Style Artist"

GENRE

MOOD

ARTIST

LYRICS GENERATOR

* Please input a lyric topic

Artist:

Language:

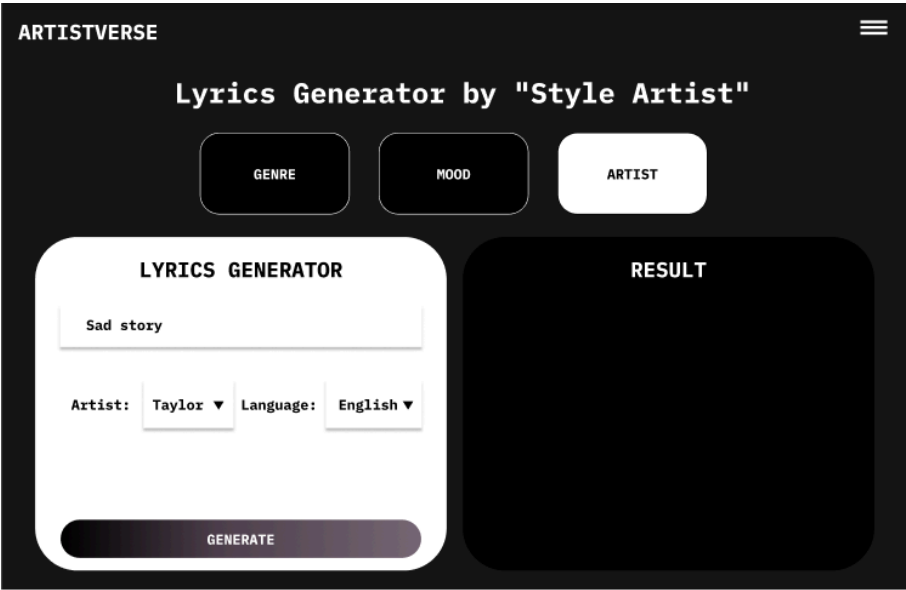
* Please select Artist * Please select language

GENERATE

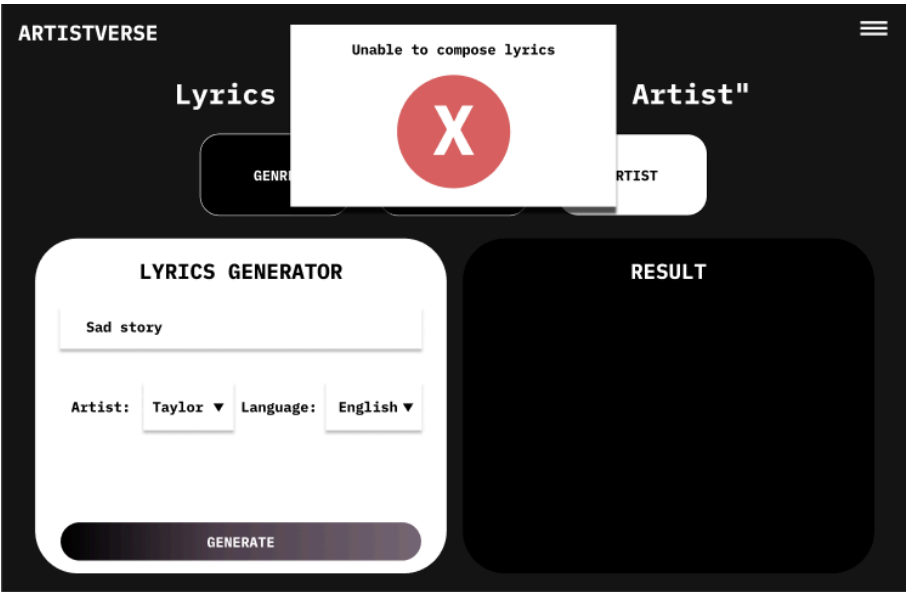
RESULT

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UI-06-2 With input information



UI-06-3 With causing by no internet connection



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UI-06-4 With display the created lyrics

ARTISTVERSE

Lyrics Generator by "Style Artist"

GENREMOODARTIST

LYRICS GENERATOR

Sad story

Artist: Taylor ▼ Language: English ▼

GENERATE

RESULT

Verse 1:
We met in the summertime, laughter in the air
Your eyes were like a sunrise, lighting up everywhere
We danced under the moonlight, whispers in the dark
I thought we had a fairytale, but you left me with a scar
Chorus:
You said forever, but you meant goodbye
Left me here in pieces, wondering why
The love we had was just a lie
Now all that's left are tears I cry

Save lyrics

UI-07 Lyric Generator page with save lyrics

ARTISTVERSE

Lyrics Generator by "Genre"

GENREMOODARTIST

Lyrics Generator

My love

Genre: Pop ▼ Language: English ▼

GENERATE

RESULT

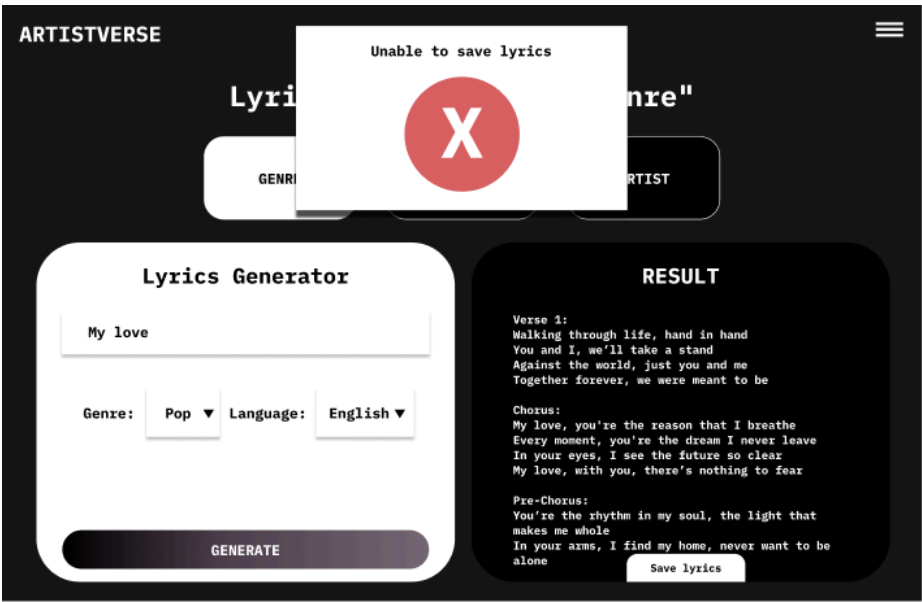
Verse 1:
Walking through life, hand in hand
You and I, we'll take a stand
Against the world, just you and me
Together forever, we were meant to be
Chorus:
My love, you're the reason that I breathe
Every moment, you're the dream I never leave
In your eyes, I see the future so clear
My love, with you, there's nothing to fear
Pre-Chorus:
You're the rhythm in my soul, the light that makes me whole
In your arms, I find my home, never want to be alone

Save lyrics

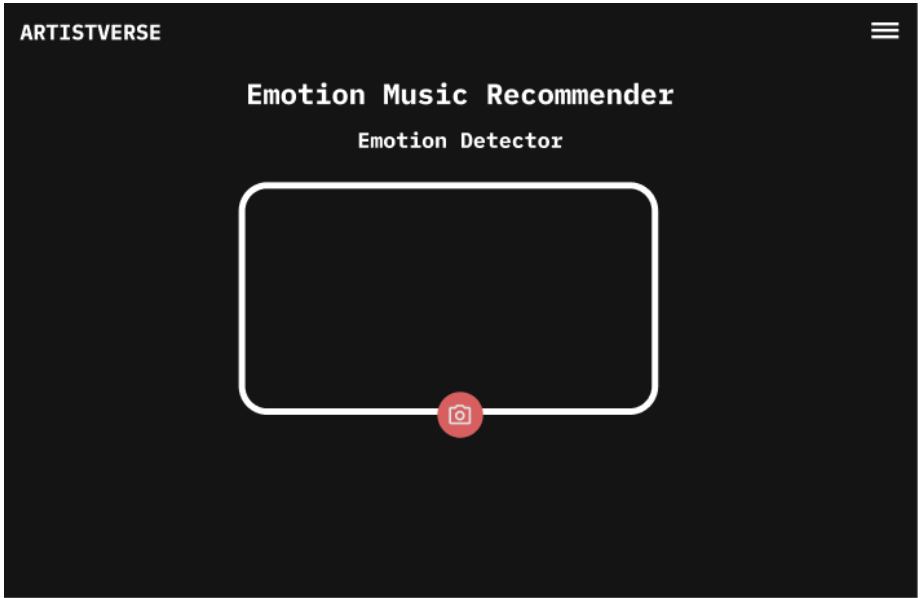
Lyrics.txt
"progress 13.23kb / 23.41kb"

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UI-07-1 With Unable to save lyrics causing by no internet connection

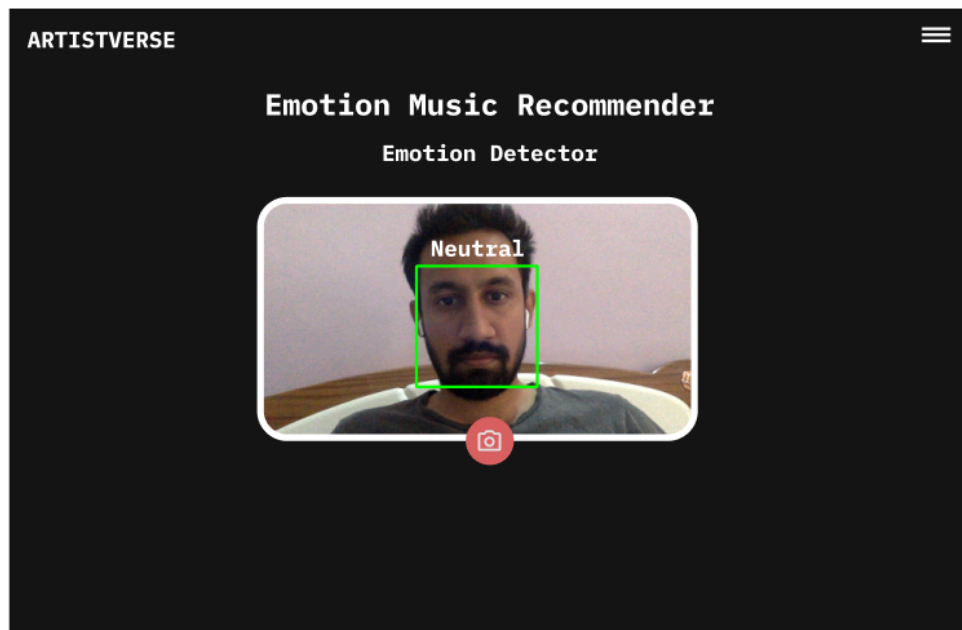


UI-08 Camera mood page

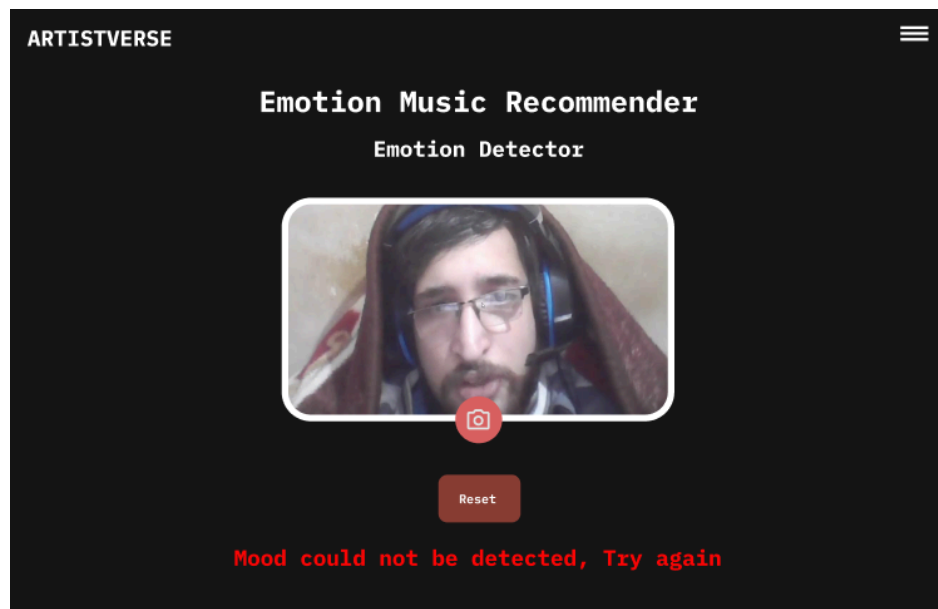


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UI-08-1 With camera view

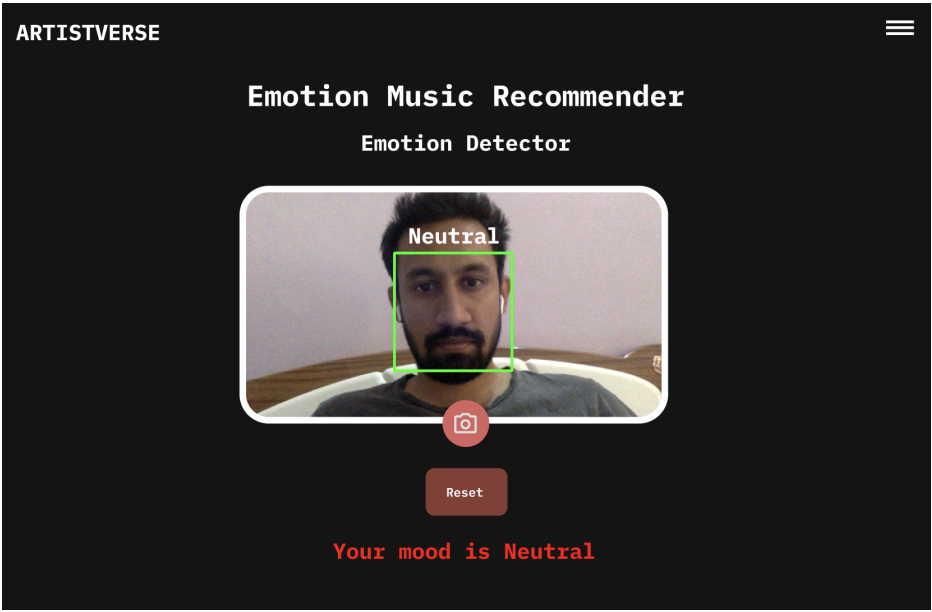


UI-08-2 With system unable to determine the mood



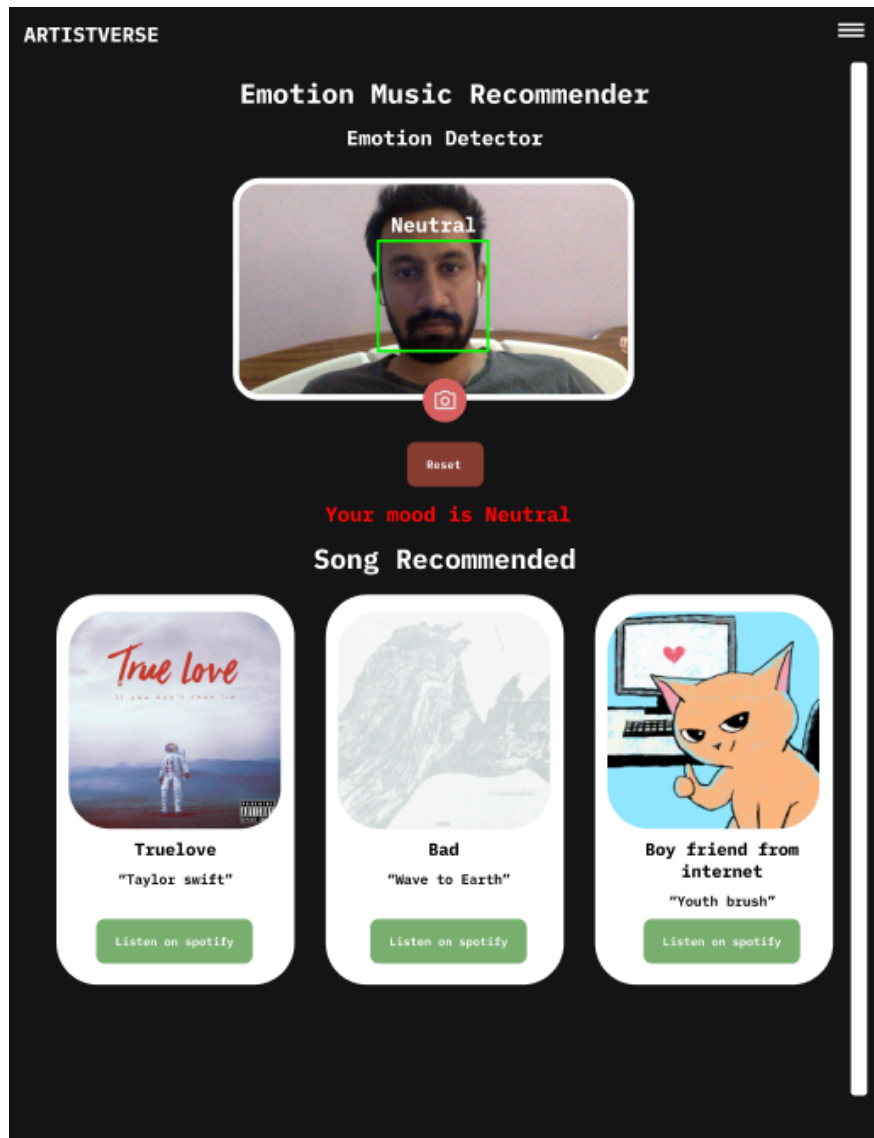
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UI-08-3 Camera mood page with detected mood



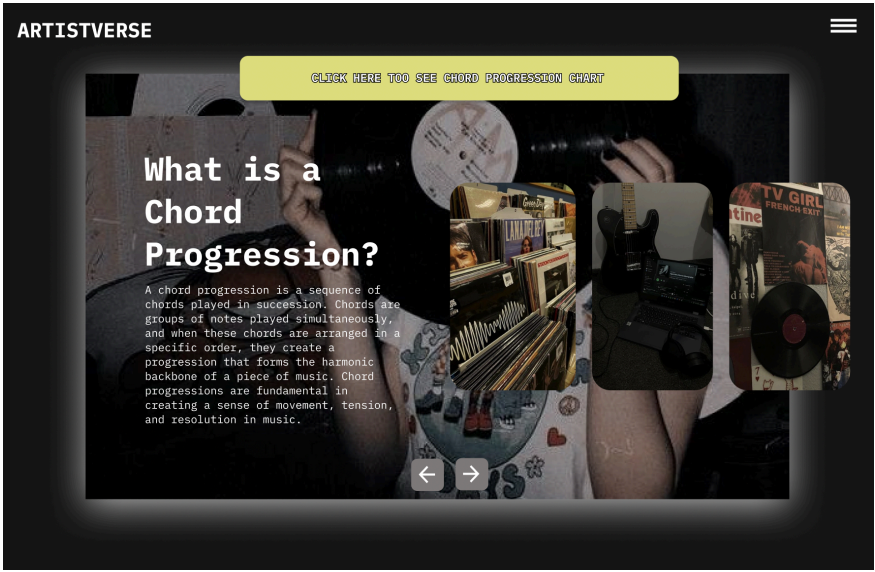
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UI-08-4 With song recommended and song choosing



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UI-09 Chord Progression page



UI-09-1 Chord Progression page when click next button



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UI-10 Chord Progression Table page

ARTISTVERSE

Major Keys Chord Table

Key	Progression								1	2	3	4	5	6	7
C	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	C	Dm	Em	F	G	Am	Bdim
D	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	D	Em	Gbm	G	A	Bm	C#dim
E	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	E	Gbm	Abm	A	B	Dbm	D#dim
F	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	F	Gm	Am	Bb	C	Dm	Edim
G	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	G	Am	Bm	C	D	Em	F#dim
A	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	A	Bm	Dbm	D	E	Gbm	G#dim
B	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	B	Dbm	Ebm	E	Gb	Abm	A#dim

UI-10-1 With selected progression

ARTISTVERSE

Major Keys Chord Table

Key	Progression								1	2	3	4	5	6	7
C	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	C	Dm	Em	F	G	Am	Bdim
D	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	D	Em	Gbm	G	A	Bm	C#dim
E	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	E	Gbm	Abm	A	B	Dbm	D#dim
F	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	F	Gm	Am	Bb	C	Dm	Edim
G	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	G	Am	Bm	C	D	Em	F#dim
A	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	A	Bm	Dbm	D	E	Gbm	G#dim
B	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	B	Dbm	Ebm	E	Gb	Abm	A#dim

Selected Chord Progression: 2-5-1-6 in Key C

Dm

G

C

Am

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UI-10-2 With highlights the chord in the progression with an orange colour as the sound it is played.

ARTISTVERSE

Major Keys Chord Table

Key	Progression								1	2	3	4	5	6	7
C	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	C	Dm	Em	F	G	Am	Bdim
D	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	D	Em	Gbm	G	A	Bm	C#dim
E	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	E	Gbm	Abm	A	B	Dbm	D#dim
F	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	F	Gm	Am	Bb	C	Dm	Edim
G	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	G	Am	Bm	C	D	Em	F#dim
A	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	A	Bm	Dbm	D	E	Gbm	G#dim
B	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	B	Dbm	Ebm	E	Gb	Abm	A#dim

Selected Chord Progression: 2-5-1-6 in Key C

G

Drop Variation 1

C

Drop Variation 1

D

Drop Variation 1

G

Drop Variation 1

UI-11 AI-Assistance page

ARTISTVERSE

Welcome to AI Assistance

This interface provides music suggestions based on lyrics you provide.

Your lyrics

Suggestion

Type a message...

Send

Save Analyzed Lyrics

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UI-11-1 With invalid input

ARTISTVERSE

☰

Welcome to AI Assistance

This interface provides music suggestions based on lyrics you provide.

Your lyrics

Suggestion

The lyrics field cannot be empty. Enter your lyrics to proceed.

Type a message...

Send

Save Analyzed Lyrics

UI-11-2 With User input

ARTISTVERSE

☰

Welcome to AI Assistance

This interface provides music suggestions based on lyrics you provide.

Your lyrics

Suggestion

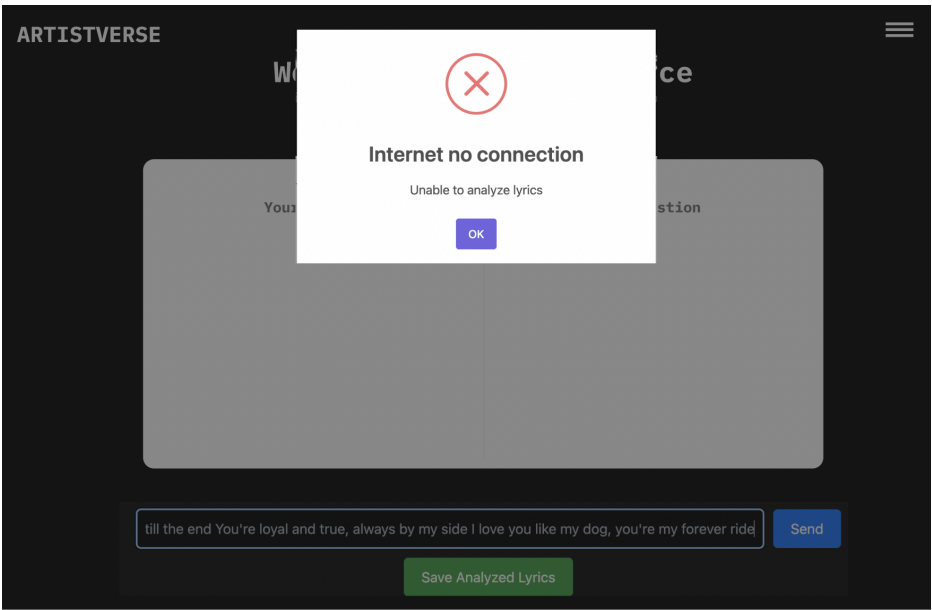
till the end You're loyal and true, always by my side I love you like my dog, you're my forever ride

Send

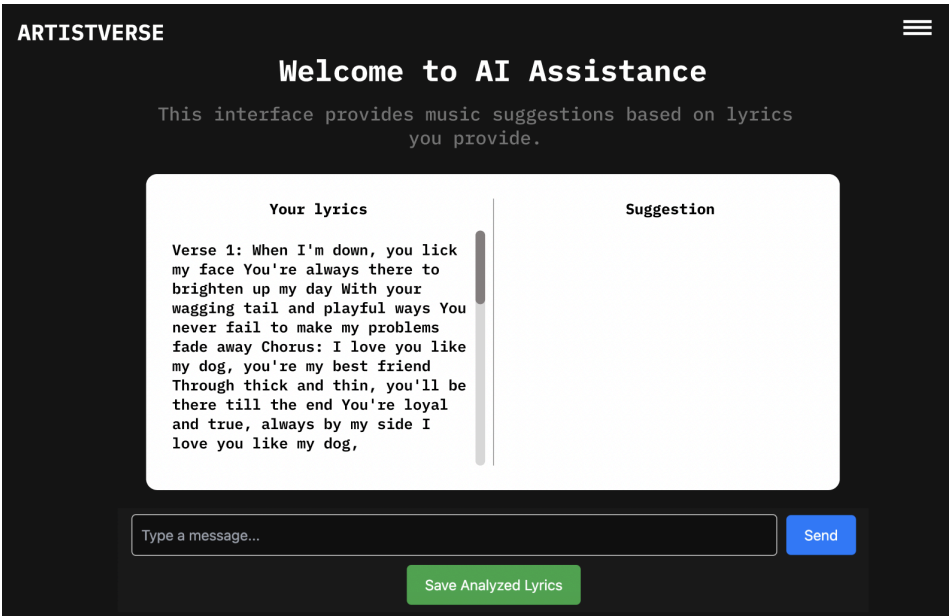
Save Analyzed Lyrics

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UI-11-3 Proceed without internet connection

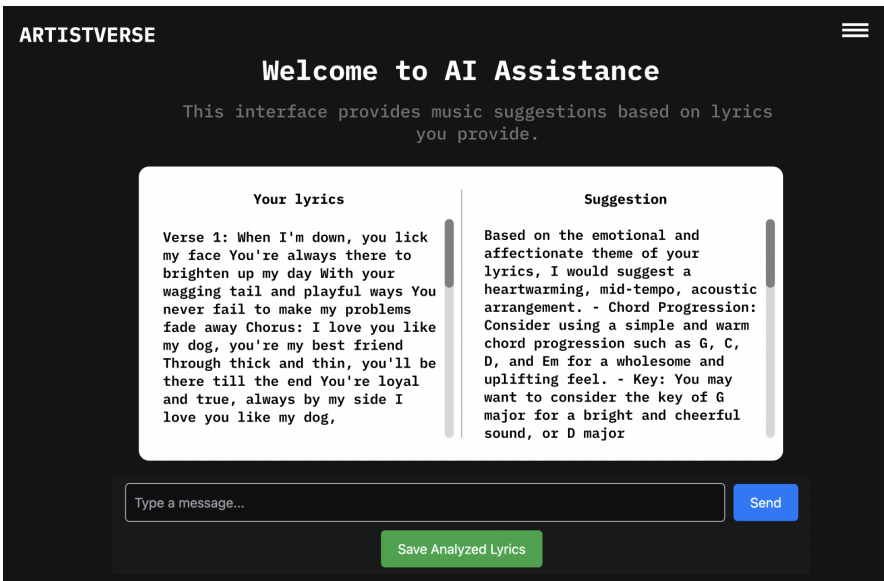


UI-11-4 With input after click button “Send”

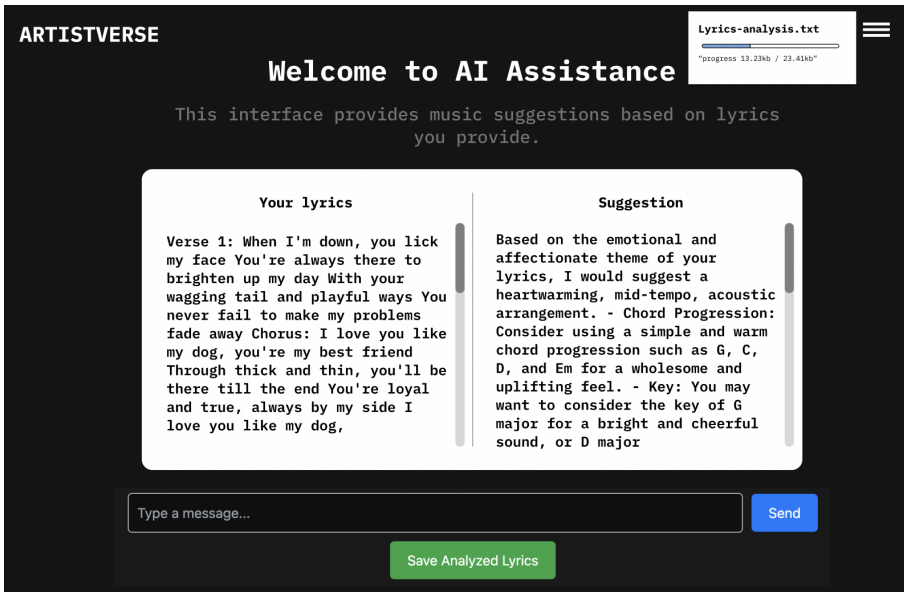


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UI-11-5 With suggestion analyzed

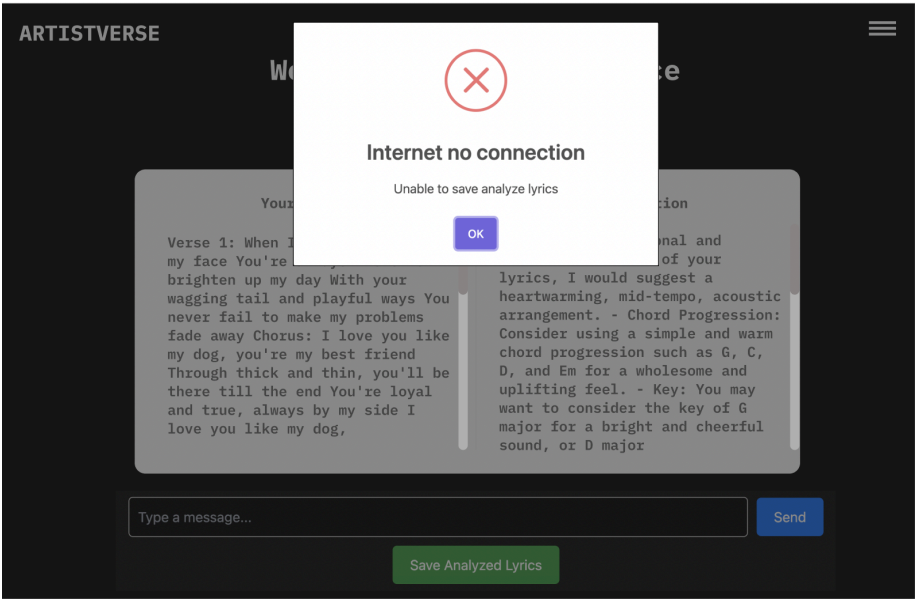


UI-11-6 With User click save analyzed lyrics



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UI-11-7 With no internet connection when clicking “Save Analyzed Lyrics”



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